Lecture 4B

資料：計量

CIE7010 交通安全設計與分析
2026年3月25日 陳彥向

2024.Mar.20 第一版，當時為 Lec 3道路安全對策評估之計量方法 (含課本第四章)

Data: Quantitative
CIE7010 Traffic Safety Design and Analysis
Mar 25, 2026
Instructor: Yen-Hsiang Chen

Background Photo: 鼎漢
Powerpoint Template Source: http://www.free-powerpoint-templates-design.com

- 有嚴謹的評估方法，交通安全的效益(effect)才是有效的(valid)
- 否則，只是簡單加減乘除，無法說服人

大綱

PART I 評估

- 評估的角色

PART II 評估的困難點

- 迴歸趨中(Regression-to-the-mean)
- 外在環境變化

PART III 計量方法基礎

- 古典最小方差(OLS)
- 推論
- 雙元變數
- 時間趨勢
- 誤差是不全知
- 線性、卜瓦松、負二項

PART IV 案例

- 免疫時間窗
- Odds Ratio
- 建模
- 車流為曝光量
- D-i-D
- 縱橫資料

來源:Google Map
@57.7276442,11.9719096

PART I: 評估

Evaluations

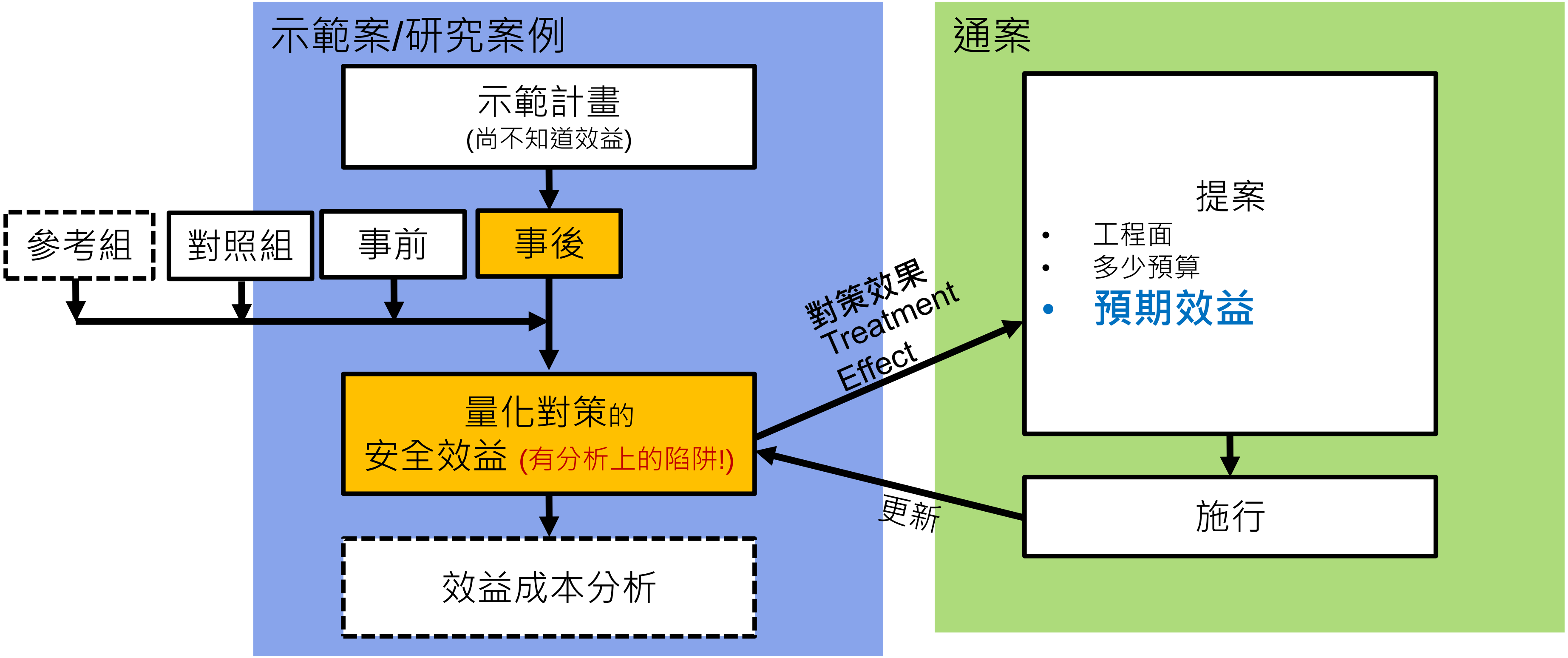
評估的角色

前言—路口安全評估

- 路口改善案，猶如頭痛醫頭、腳痛醫腳
  - ※路網才能看到全貌

- 但至少仍要知道如何醫頭、醫腳

量化安全效益


示範案/研究案例
示範計畫
(尚不知道效益)
參考組
對照組
事前
事後
量化對策的
安全效益 (有分析上的陷阱!)
效益成本分析
對策效果
Treatment
Effect
更新
通案
提案
• 工程面
• 多少預算
• 預期效益
施行

來源:Google Map
@57.7276442,11.9719096

PART II: 評估的困難點

什麼是對策真正的效益

- 量化各改善方案的「對策效果(treatment effect)」是一件困難的事
  - 不是簡單的事前事後加減乘除
  - 要分離三個造成事故下降的來源

回歸趨中 | 大趨勢 | 對策效果

  - 大趨勢：
    - 例如：可能是大趨勢原本就事故率向下，而不是本對策之效果
    - 例如：對策可能有效，但大趨勢是事故大幅上漲而蓋過了對策

(續)

- Elvik(2002)整理了挪威各種對策，以及各種因素造成事故率變化：

R. Elvik / Accident Analysis and Prevention 34 (2002) 631–635 633

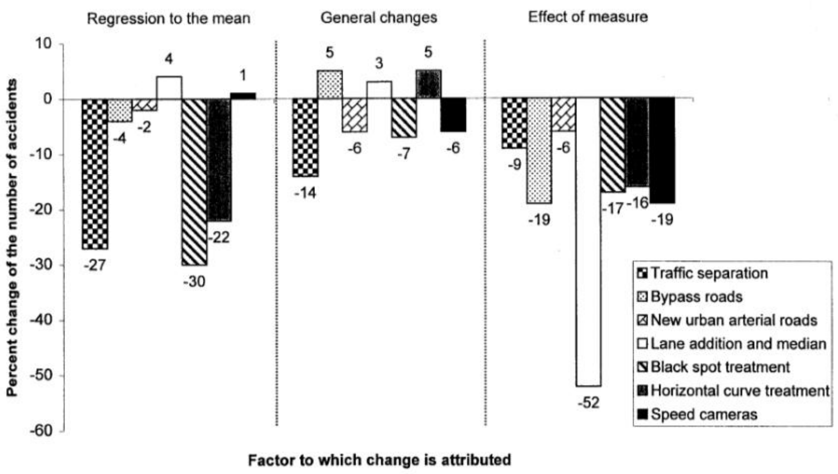


Fig. 1. Percentage change of the number of accident attributed to regression-to-the-mean, general changes and the road safety measure for seven road safety measures.

(續)

- 挪威增設測速照相
  - 單純的(naïve)數據事故**下降24%** (非正確答案!)
    - 用簡單的加減乘除
  - 但實質上是**下降19%的效果**
    - 挪威當年大趨勢是事故一直下降(達6%)
    - 分離「回歸趨中」、「大趨勢」後，才能得知真正的效益!

單純的
(-24%)

回歸趨中
+1%

大趨勢
(-6%)

對策效果
(-19%)

需要處理的議題

Issues Need to be Addressed

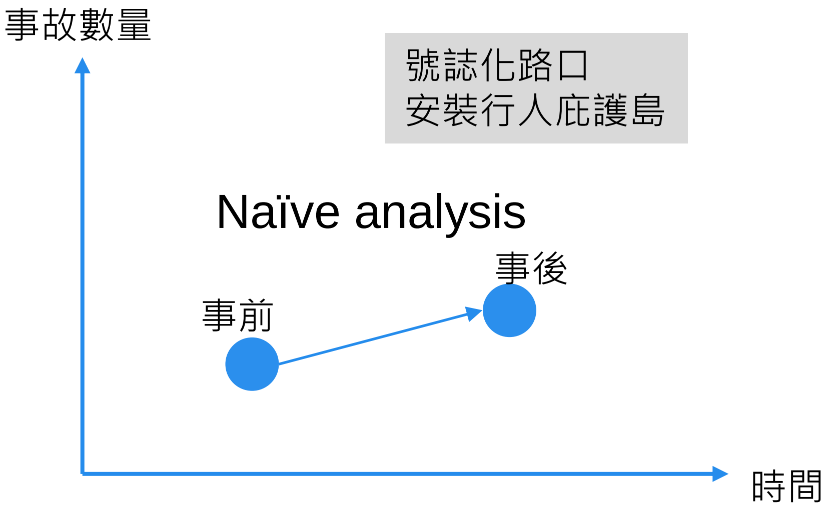
所以，有效(valid)的量化分析，須解決

- 1. **回歸趨中**(Regression-to-the-mean, RTM)
    - Selection bias
    - Not sampling average

- 2. **時間趨勢效應**(Time trend effects) (El-Basyouny & Sayed, 2011; Guo & Sayed, 2020)

大趨勢

樸素的(Naïve)分析不可盡信

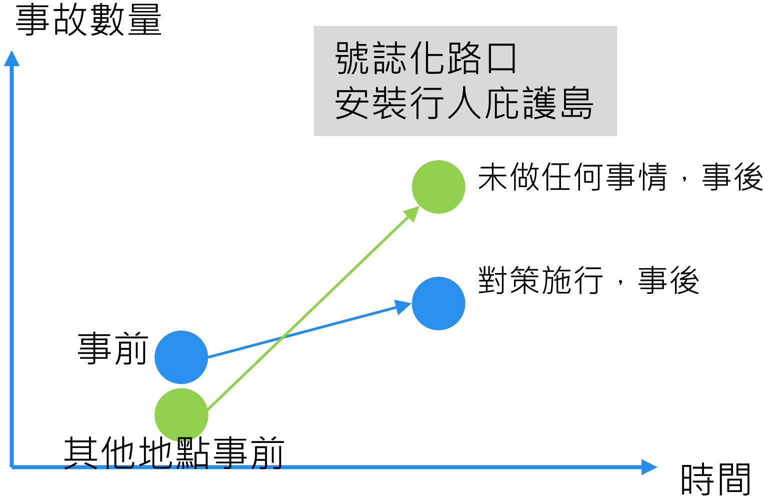


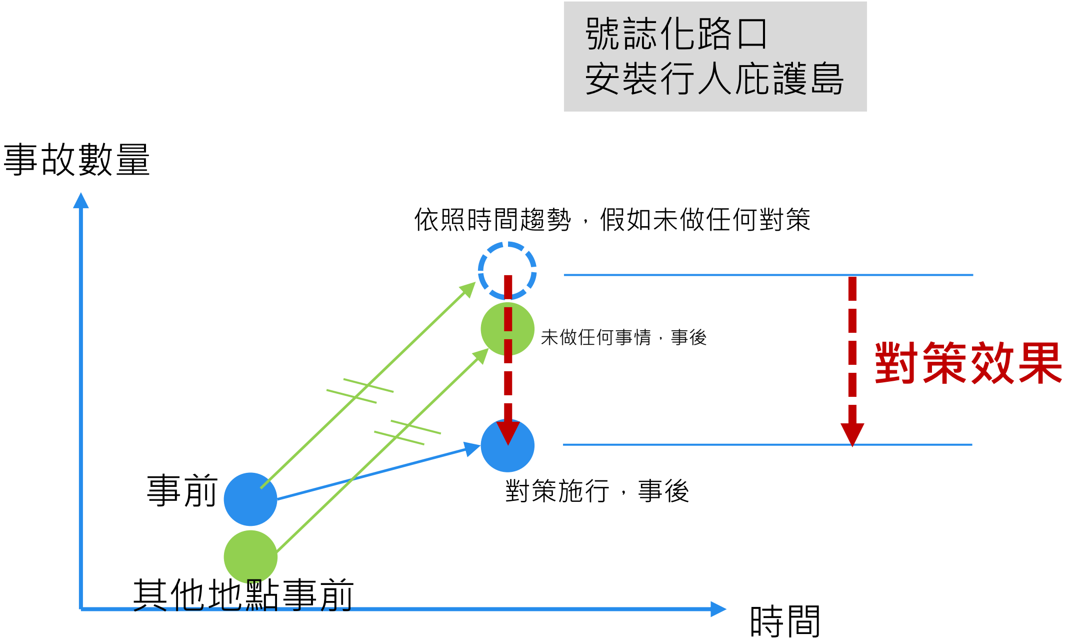
用加減乘除
隨意下錯誤結論

「*安裝行人庇護島無效*」

「時間趨勢效應」

Time Trend Effects





迴歸趨中

(Regression-to-the-mean)

隨機性

- 雖然平均固定，但上下起伏有隨機性
- 假如某個設施，每個地點、每年事故數量平均為3.6起，卜瓦松分布(Poison distributed)
  - 可能的結果為(由EXCEL隨機產生)：

	T(-2)	T(-1)	T0	T1	T2
Site1	5	1	5	3	5
Site2	5	5	2	1	5
Site3	3	4	4	2	3
Site4	6	5	5	5	2
Site5	7	3	8	2	3
Site6	1	5	5	0	5
Site7	3	4	0	2	6
Site8	4	2	5	6	3
Site9	6	0	0	4	1
Site10	1	2	4	3	3
Site11	2	4	3	4	6
Site12	4	6	4	2	3
Site13	2	0	0	2	4
Site14	1	7	3	6	5
Site15	2	8	3	1	2
Site16	1	3	5	0	2
Site17	3	2	5	4	4
Site18	2	1	1	4	4
Site19	1	9	6	2	1
Site20	3	2	2	1	1

(續)

- 假如作了某對策，例如掛「易肇事路口，請小心行駛」的標誌，假如完全沒有效果，那麼事前、事後都維持 3.6事故/年 的平均值，卜瓦松分布。
  – 取5個地點，事前、事後 單純(naïve)對比：誤以為降低事故 45.8%!

Few sites with few periods				
	T0	T1		
Site1	5	3		
Site2	2	1		
Site3	4	2		
Site4	5	5		
Site5	8	2		
Average	4.8	2.6		-45.8%

(續)

- 若取更多地點(增至10個)，誤以為降26.3%(但稍微好一點)：

More Sites				
	T0	T1		
Site1	5	3		
Site2	2	1		
Site3	4	2		
Site4	5	5		
Site5	8	2		
Site6	5	0		
Site7	0	2		
Site8	5	6		
Site9	0	4		
Site10	4	3		
	3.8	2.8		-26.3%

(續)

- 若取更多時段(事前增至2年、事後增至2年) ，誤以為降26.2%(但稍微好一點) ：

	More Periods				
	T(-1)	T0	T1	T2	
Site1	1	5	3	5	
Site2	5	2	1	5	
Site3	4	4	2	3	
Site4	5	5	5	2	
Site5	3	8	2	3	
Average	4.2		3.1		-26.2%

(續)

- 若時間、地點等樣本都蒐集較多→誤以為對策降低事故7.2%：

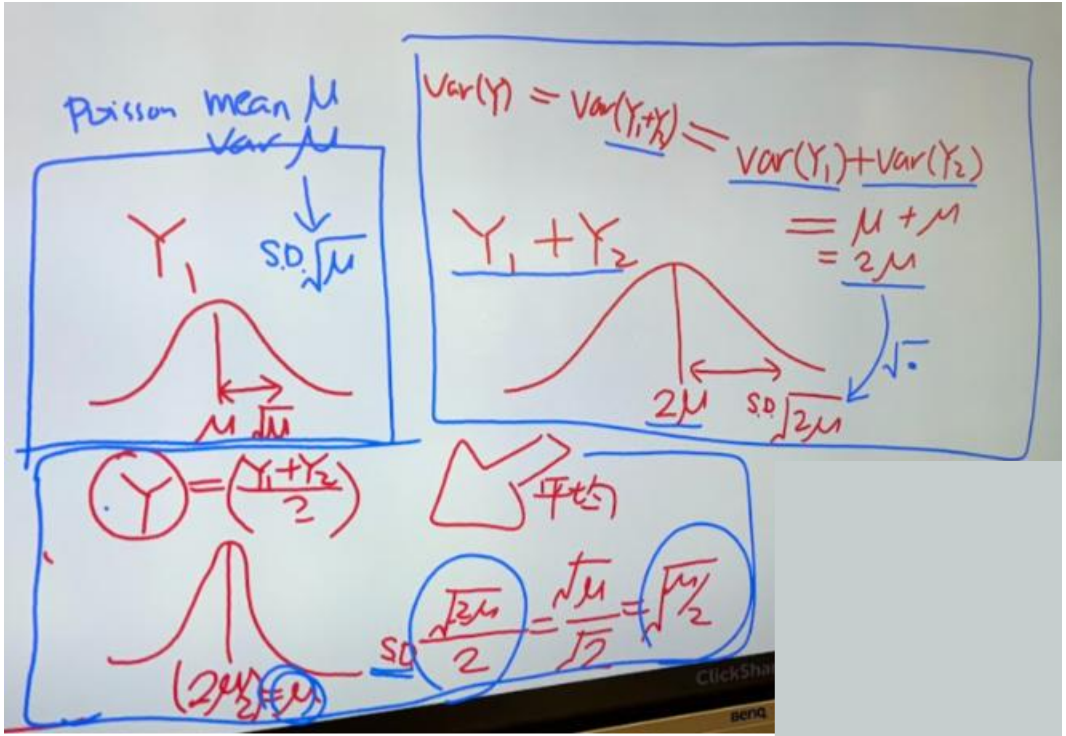
	MorePeriods with More Sites				
	T(-1)	**T0**	**T1**	**T2**	
Site1	1	5	3	5	
Site2	5	2	1	5	
Site3	4	4	2	3	
Site4	5	5	5	2	
Site5	3	8	2	3	
Site6	5	5	0	5	
Site7	4	0	2	6	
Site8	2	5	6	3	
Site9	0	0	4	1	
Site10	2	4	3	3	
Average	3.45		3.2		-7.2%

隨機性而造成的RTM

- 隨機性而造成的RTM → **樣本變多可以舒緩**
  - 事前偏高於平均，事後自然較可能會靠近平均—單純的(naively)看起來改進
  - 事前偏低於平均，事後自然較可能會靠近平均—單純的(naively)看起來惡化

- 理論上：
  - Y1~ Poisson with mean (μ) → Var = μ , standard deviation $\sqrt{\mu}$
  - Y2~ Poisson with mean (μ) → Var = μ , standard deviation $\sqrt{\mu}$
  - Z = Y1+Y2, Z~ Poisson with mean (2 mu) → Var (2 μ)
  - Y = Z/2 Y~NON-Poisson with mean (mu), standard deviation
    - $\left(\frac{1}{2}\right)\sqrt{2\mu} = \sqrt{\mu/2}$

- 那麼，有n 個Poisson相加：
    - Z = $\sum_{i=1}^{N} Y_i$ ~ Poisson ($N\ \mu$) → Mean ($N\ \mu$) 、Var($N\ \mu$) → standard deviation $\sqrt{N\mu}$
    - Z仍服從Poisson
        - Sanity check: (Var/Mean) = 1
- 取平均：
    - Y = (1/N) $\sum_{i=1}^{N} Y_i$ → Mean μ , standard deviation $\sqrt{\mu/N}$
    - Y不是Poisson: (Var/Mean) = 1/N
- 若取無限個樣本
    - Y → Mean μ, standard deviation 0: completely wipe out the randomness from R-T-M


Poisson mean μ
Var μ
S.D. √μ
Y1
μ √μ
Var(Y) = Var(Y1+Y2) = Var(Y1)+Var(Y2)
= μ+μ
= 2μ
Y1 + Y2
2μ
S.D √2μ
Y = (Y1+Y2)/2
平均
(2μ/2) = μ
SD √2μ/2 = √μ/√2 = √(μ/2)

事前、事後

現在，討論不限於Poisson，而是任何一種Random variable

- 假如改善確實有效果
  - 事前 Y_i~(μ , σ_B^2) , M個樣本 $Y^B = (\frac{1}{M}) \sum_{i=1}^{M} Y_i$
  - 事後 Y_j ~(ν ,σ_A^2) , N個樣本 $Y^A = (\frac{1}{N}) \sum_{i=1}^{N} Y_j$
- 事前的Random Variable機率分布：
  - $\sum_{i=1}^{M} Y_i$ 有平均$M\mu$, 變異數$M\sigma_B^2$
  - $\sum_{i=1}^{M} Y_i$ 有平均$M\mu$, 標準差$\sigma_B \cdot \sqrt{M}$
  - $Y^B = (\frac{1}{M}) \sum_{i=1}^{M} Y_i$有平均$\mu$，標準差$\sqrt{M}$ $\frac{\sigma_B \sqrt{M}}{M} = \sigma_B / \sqrt{M}$

- 同理，事後的Random Variable機率分布：
  - $Y^A = (\frac{1}{N}) \sum_{j=1}^{N} Y_j$有平均$\nu$，標準差$\frac{\sigma_A \sqrt{N}}{N} = \sigma_A / \sqrt{N}$

$Y^B = (\frac{1}{M})\sum_{i=1}^{M} Y_i$有平均$\mu$，標準差$\frac{\sigma_B\sqrt{M}}{M} = \frac{\sigma_B}{\sqrt{M}}$

$Y^A = (\frac{1}{N})\sum_{j=1}^{N} Y_j$有平均$\nu$，標準差$\frac{\sigma_A\sqrt{N}}{N} = \sigma_A/\sqrt{N}$

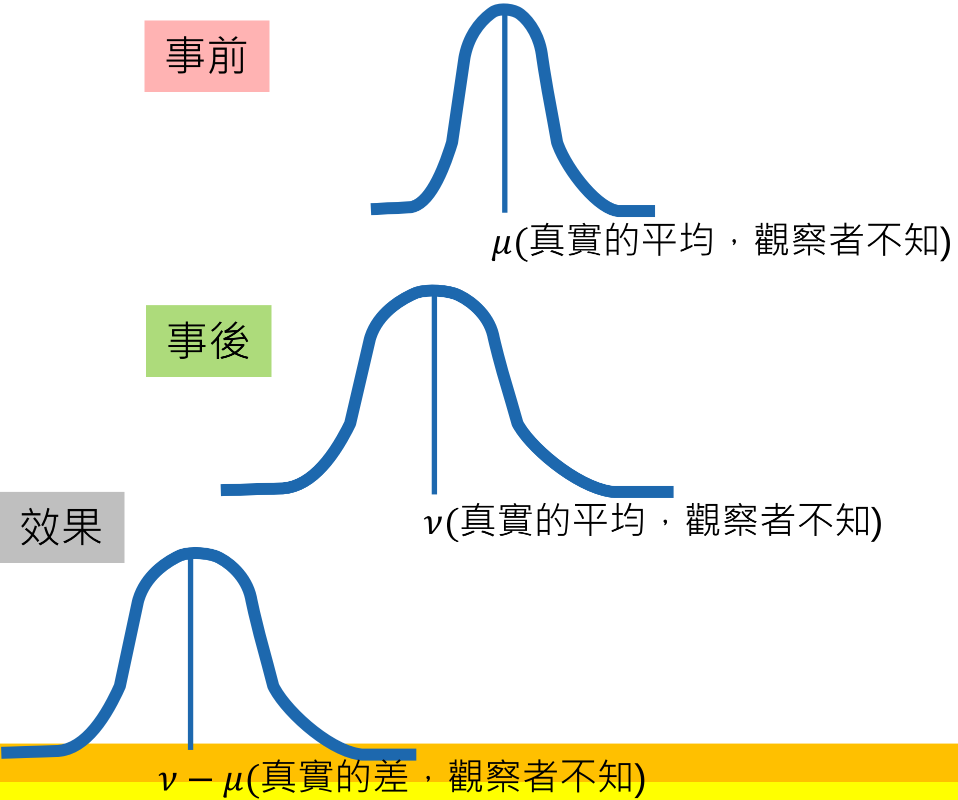
(續)

- 對策效果，事後減去事前
  - $(Y^A - Y^B)$
  - 沒有辦法直接計算 兩個RV相減(除非假設特殊的分布例如Poisson)
  - 如果樣本夠多$\sum_{i=1}^{M} Y_i$接近常態分布
    - » Y^B 也接近常態分布→ N(μ, $\left(\frac{\sigma_B}{\sqrt{M}}\right)^2$)
  - 如果樣本夠多$\sum_{j=1}^{N} Y_j$接近常態分布
    - » Y^A也接近常態分布→ N(ν, $\left(\frac{\sigma_A}{\sqrt{N}}\right)^2$)
  - 常態分佈可以相減，仍為常態分佈
    - » 所以 $(Y^A - Y^B)$ ~ N($\nu - \mu$, $\left(\frac{\sigma_B}{\sqrt{M}}\right)^2 + \left(\frac{\sigma_A}{\sqrt{N}}\right)^2$)

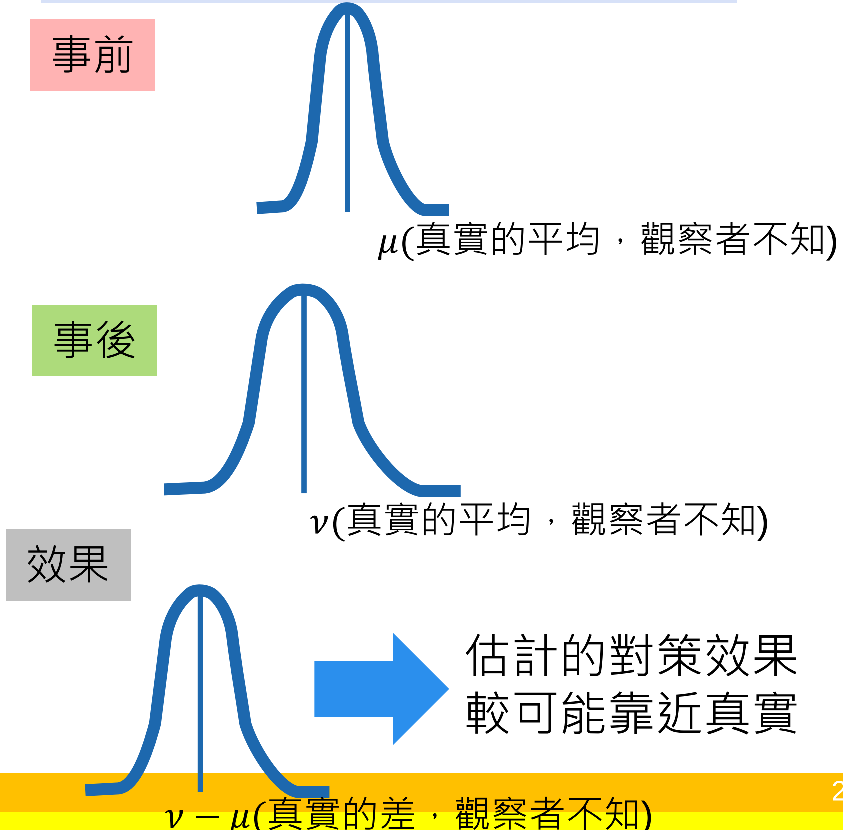
(續)

- 圖形化的概念為：

樣本事前n個、事後m個



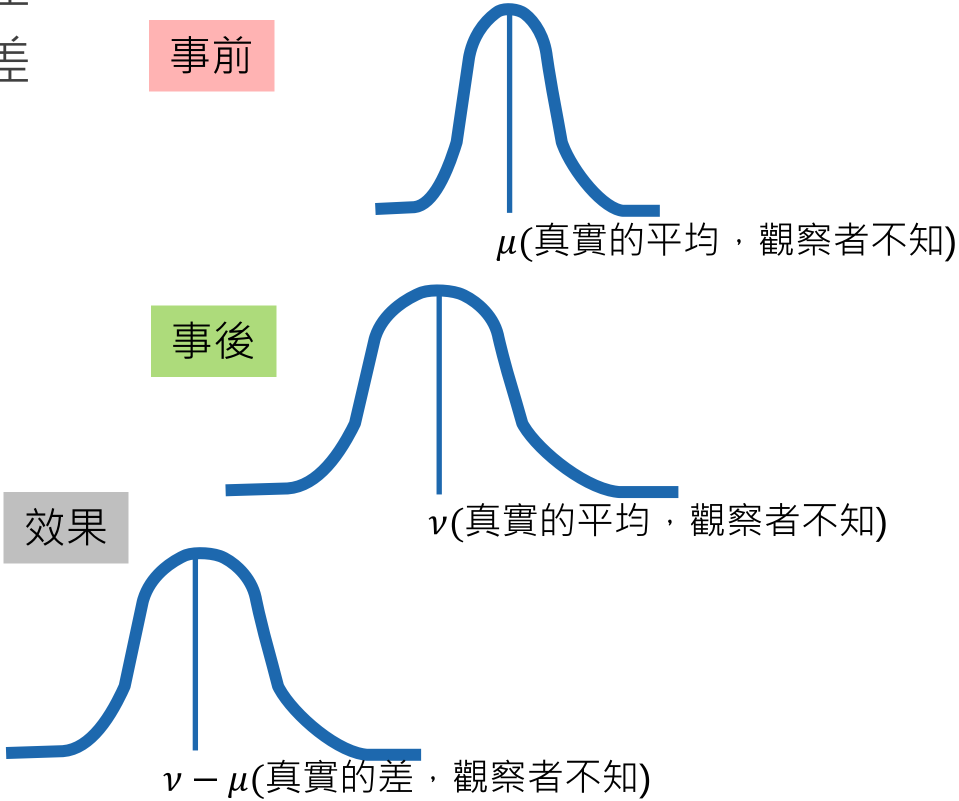
樣本事前N個、事後M個
N>n
M>m



練

(續)

- 1個標準差
- 2個標準差



選擇偏見

例

Poisson(mu = 3.6)

- 50個位址、5期

	T(-2)	T(-1)	T0	T1	T2		BEFORE(3 periods)
Site1	*2*	*2*	1	3	6		1.67
Site2	*4*	*0*	1	8	3		1.67
Site3	*7*	7	6	6	3		6.67
Site4	*2*	5	4	2	4		3.67
Site5	*6*	6	7	8	8		6.33
Site6	*0*	3	4	1	1		2.33
Site7	*2*	6	5	5	2		4.33
Site8	*1*	2	2	5	1		1.67
Site9	*3*	6	4	3	3		4.33
Site10	*2*	5	7	6	5		4.67
Site11	*4*	8	4	6	2		5.33
Site12	*1*	1	3	3	5		1.67
Site13	*1*	7	3	5	2		3.67
Site14	*10*	3	1	1	4		4.67
Site15	*4*	4	3	6	5		3.67
Site16	*3*	3	2	3	3		2.67
Site17	*4*	3	2	3	5		3.00
Site18	*3*	3	1	3	1		2.33
Site19	1	0	1	2	3		0.67
Site20	3	0	1	5	1		1.33
Site21	1	5	8	7	4		4.67
Site22	3	2	3	2	5		2.67
Site23	1	6	4	5	4		3.67
Site24	2	3	4	6	5		3.00
Site25	3	3	6	4	3		4.00
Site26	5	1	4	5	1		3.33
Site27	4	5	8	2	5		5.67
Site28	2	4	5	5	2		3.67
Site29	5	2	1	2	4		2.67
Site30	6	2	1	5	4		3.00
Site31	4	2	4	2	5		3.33
Site32	2	6	2	3	7		3.33
Site33	5	5	4	2	6		4.67
Site34	5	3	4	2	4		4.00
Site35	3	7	5	4	4		5.00
Site36	5	3	7	2	3		5.00
Site37	1	4	2	1	4		2.33
Site38	7	3	7	3	4		5.67
Site39	3	2	2	3	3		2.33
Site40	2	2	4	2	6		2.67
Site41	4	3	4	3	2		3.67
Site42	4	6	3	4	5		4.33
Site43	4	2	6	4	1		4.00
Site44	3	2	5	1	8		3.33
Site45	4	5	3	1	7		4.00
Site46	3	2	4	4	5		3.00
Site47	5	3	3	2	7		3.67
Site48	*4*	3	4	4	1		3.67
Site49	*2*	4	3	4	3		3.00
Site50	*8*	3	8	3	4		6.33
AVG	**3.46**	**3.54**	**3.80**	**3.62**	**3.86**		

- 在T0時，挑選20個表面上看來「最嚴重」的路口(跨3期)來改善’
  - → 其實是有選擇偏見! 選了本來就高於平均的路口們
  - 以下展示排名前20事故之路口

SELECTION- potential bias						4.87		3.90	Naïve Comparison -19.9%
	T(-2)	T(-1)	T0	T1	T2	BEFORE(3 periods)	Rank	AFTER(2 periods)	
Site3	7	7	6	6	3	6.67	1	4.50	
Site5	6	6	7	8	8	6.33	2	8.00	
Site50	8	3	8	3	4	6.33	3	3.50	
Site27	4	5	8	2	5	5.67	4	3.50	
Site38	7	3	7	3	4	5.67	5	3.50	
Site11	4	8	4	6	2	5.33	6	4.00	
Site35	3	7	5	4	4	5.00	7	4.00	
Site36	5	3	7	2	3	5.00	8	2.50	
Site10	2	5	7	6	5	4.67	9	5.50	
Site14	10	3	1	1	4	4.67	10	2.50	
Site21	1	5	8	7	4	4.67	11	5.50	
Site33	5	5	4	2	6	4.67	12	4.00	
Site7	2	6	5	5	2	4.33	13	3.50	
Site9	3	6	4	3	3	4.33	14	3.00	
Site42	4	6	3	4	5	4.33	15	4.50	
Site25	3	3	6	4	3	4.00	16	3.50	
Site34	5	3	4	2	4	4.00	17	3.00	
Site43	4	2	6	4	1	4.00	18	2.50	
Site45	4	5	3	1	7	4.00	19	4.00	
Site4	2	5	4	2	4	3.67	20	3.00	

誤以為改善19.9%

小結

時間趨勢與消除方法

- 同時有事前事後比較(before-and-after-comparison)和對照組(comparison group)

RTM

Sample size

Selection bias

trend

回歸趨中與消除方法

「回歸趨中(R-T-M)」只是一個現象，本身並不是問題

- (1)樣本太少才是問題
- (2)選擇偏誤(selection bias)才是問題

同時消除R-T-M之「自然取樣偏誤」與「選擇偏見」：

- 經驗貝氏法(Empirical Bayesian, EB)：
  - 3組資料：事前事後組、對照組、參考組(reference group)
- 全貝氏法(Full Bayesian, FB)

消除R-T-M之自然取樣偏誤：

- 擴大樣本 (壓低 $\sigma/\sqrt{n}$) (Wei et al., Under Review)

消除R-T-M之選擇偏誤：

- 「偏誤免疫時間窗」 (Wei et al., Under Review)

來源:Google Map
@57.7276442,11.9719096

PART III: 計量方法基礎

- 諾貝爾獎：

nobelprize.org/prizes/economic-sciences/2021/angrist/facts/

Economic Sciences | Prize in Economic Sciences 2021 | Joshua D. Angrist - Facts

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

David Card
Joshua D. Angrist
Guido W. Imbens

Share this

Joshua D. Angrist
Facts

© Nobel Prize Outreach. Photo: Stefan Bladh

Joshua D. Angrist
The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

Born: 18 September 1960, Columbus, OH, USA

Affiliation at the time of the award: Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Prize motivation: “for their methodological contributions to the analysis of causal relationships”

Prize share: 1/4

古典最小方差

Ordinary Least Square (OLS)

回歸是什麼?

找尋應變數(Y)的可能構成——由自變數(independent variable, ***covariates***)組合成的因素

原始資料

就學時長(年)	收入
Edu	Y
12	148
18	111
18	159
18	140
18	194
22	135
12	85
6	63
12	135
12	114
12	115
6	71
9	94
12	91
18	125

R語言 https://www.statskingdom.com/410multi_linear_regression.html

```
 if(!"car" %in% installed.packages()){install.packages("car")}
library("car")
y<-c(148,111,159,140,194,135,85,63,135,114,115,71,94,91,125)
edu<-c(12,18,18,18,18,22,12,6,12,12,12,6,9,12,18)
model1 = lm(y~edu)
```

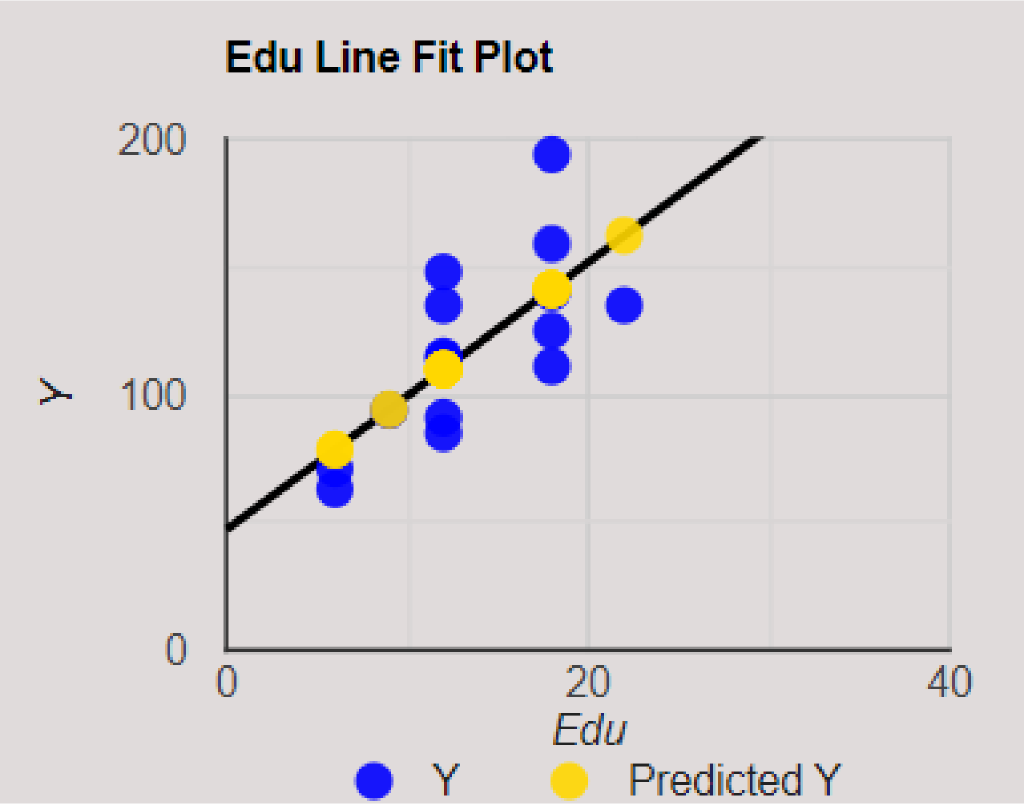
Model 1:

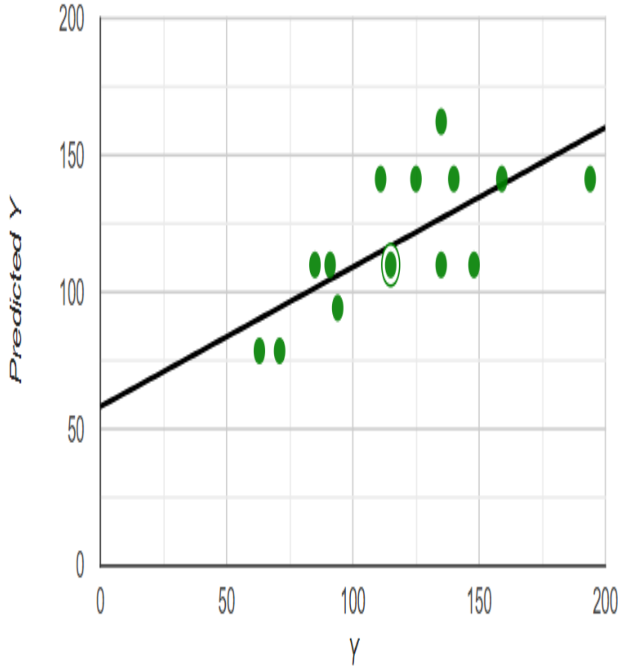
$$\hat{Y} = 47.1378 + 5.2338\ Edu$$

「年薪和書讀得越久正相關
每多讀一年書年薪多5.233萬」

R square (R^2) equals **0.5097**

Adjusted R square equals **0.4719**


Edu Line Fit Plot
200
100
0
Y
0
20
40
Edu
Y
Predicted Y


200
150
100
50
0
Predicted Y
0
50
100
150
200
Y

能否有效預測?

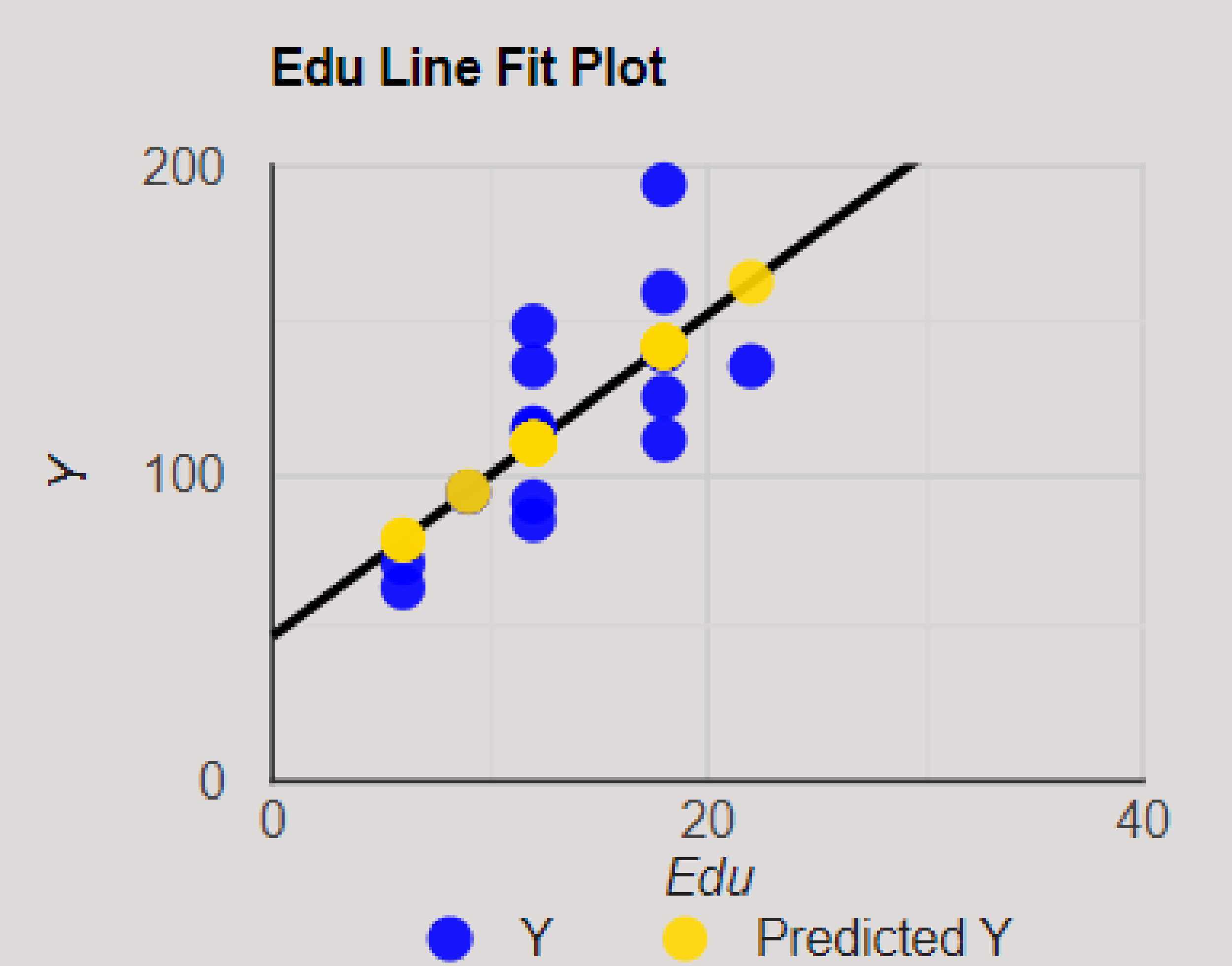
- 不能完全預測。必定有偏誤(error)
  - Y-hat → 假如按照回歸式，由各個X推得
  - Y → 真正觀察到的值

(Model 1)

$\hat{Y}$ = 47.1378 + 5.2338 Edu

Y	Y-hat	Error
148	110.08	37.9
111	141.54	-30.5
159	141.54	17.5
140	141.54	-1.5
194	141.54	52.5
135	162.52	-27.5
85	110.08	-25.1
63	78.607	-15.6
135	110.08	24.9
114	110.08	3.9
115	110.08	4.9
71	78.607	-7.6
94	94.341	-0.3
91	110.08	-19.1
125	141.54	-16.5

i	Y	Y-hat	Error
1	148	110.08	37.9
2	111	141.54	-30.5
3	159	141.54	17.5
4	140	141.54	-1.5
5	194	141.54	52.5
6	135	162.52	-27.5
7	85	110.08	-25.1
8	63	78.607	-15.6
9	135	110.08	24.9
10	114	110.08	3.9
11	115	110.08	4.9
12	71	78.607	-7.6
13	94	94.341	-0.3
14	91	110.08	-19.1
15	125	141.54	-16.5



- (續)

(Model 1)

$\hat{Y}$ ***= 47.1378 + 5.2338 Edu***

Y	Y-hat	Error	Error
148	110.08	37.9	ε_1 = 37.9
111	141.54	-30.5	ε_2 = -30.5
159	141.54	17.5	ε_3 = 17.5
140	141.54	-1.5	ε_4 = -1.5
194	141.54	52.5	ε_5 = 52.5
135	162.52	-27.5	ε_6 = -27.5
85	110.08	-25.1	ε_7 = -25.1
63	78.607	-15.6	ε_8 = -15.6
135	110.08	24.9	ε_9 = 24.9
114	110.08	3.9	ε_{10} = 3.9
115	110.08	4.9	ε_{11} = 4.9
71	78.607	-7.6	ε_{12} = -7.6
94	94.341	-0.3	ε_{13} = -0.3
91	110.08	-19.1	ε_{14} = -19.1
125	141.54	-16.5	ε_{15} = -16.5

$(\varepsilon_1)^2 + (\varepsilon_2)^2 + \ldots = 8416.3$

(續)

$\hat{Y}$ = 47.1378 + 5.2338 Edu

完整的型式

- $y_i = \beta_0 \ + \beta_1 x_{i,1} + \varepsilon_\mathrm{i}$

用hat來表達型式

- $\hat{y}_i = \beta_0 \ + \beta_1 x_{i,1}$

- *(Model 1)*
- $\hat{Y}$ *= 47.1378 + 5.2338 Edu*

$(\varepsilon_1)^2 + (\varepsilon_2)^2 + \ldots = 8{,}416.3$

- *(Model 101)*
- $\hat{Y}$ *= 50 + 4 Edu*

$(\varepsilon_1)^2 + (\varepsilon_2)^2 + \ldots = 11{,}482.00$

- *(Model 201)*
- $\hat{Y}$ *= 45 + 6 Edu*

$(\varepsilon_1)^2 + (\varepsilon_2)^2 + \ldots = 9{,}645.0$

從R-code 產出的回歸式，他的誤差真的是最小的

理論解

純量式

- Min $\sum_i \varepsilon_i^2$
- Min $\sum_i \left[y_i - \sum_{j=0} \beta_j \, x_{i,j}\right]^2$

HW

理論解

純量式

- Min $\sum_i \varepsilon_i^2$
- Min $\sum_i \left[y_i - \Sigma_{j=0}\, \beta_j\, x_{i,j}\right]^2$

高中數學課本 SSx, Ssy,…

向量式

- $\min\limits_{\beta} \quad \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}$
- $\min\limits_{\beta} (\boldsymbol{y} - \boldsymbol{\beta}^T \boldsymbol{x})^2$
- F.O.C.: $/\, \partial \boldsymbol{\beta} = 0$
- 公式(向量式) : $\widehat{\boldsymbol{\beta}} = (\boldsymbol{x}^T \boldsymbol{x})^{-1} (\boldsymbol{x}^T \boldsymbol{y})$

Source: Wiki

$\mathbf{a}$ and $\mathbf{b}$ are not functions of $\mathbf{X}$	$\frac{\partial \mathbf{a}^\top \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} =$	$\mathbf{b}\mathbf{a}^\top$	$\mathbf{a}\mathbf{b}^\top$
$\mathbf{a}$ and $\mathbf{b}$ are not functions of $\mathbf{X}$	$\frac{\partial \mathbf{a}^\top \mathbf{X}^\top \mathbf{b}}{\partial \mathbf{X}} =$	$\mathbf{a}\mathbf{b}^\top$	$\mathbf{b}\mathbf{a}^\top$
$\mathbf{a}$, $\mathbf{b}$ and $\mathbf{C}$ are not functions of $\mathbf{X}$	$\frac{\partial (\mathbf{X}\mathbf{a}+\mathbf{b})^\top \mathbf{C} (\mathbf{X}\mathbf{a}+\mathbf{b})}{\partial \mathbf{X}} =$	$\left(\left(\mathbf{C}+\mathbf{C}^\top\right)(\mathbf{X}\mathbf{a}+\mathbf{b})\mathbf{a}^\top\right)^\top$	$\left(\mathbf{C}+\mathbf{C}^\top\right)(\mathbf{X}\mathbf{a}+\mathbf{b})\mathbf{a}^\top$
$\mathbf{a}$, $\mathbf{b}$ and $\mathbf{C}$ are not functions of $\mathbf{X}$	$\frac{\partial (\mathbf{X}\mathbf{a})^\top \mathbf{C} (\mathbf{X}\mathbf{b})}{\partial \mathbf{X}} =$	$\left(\mathbf{C}\mathbf{X}\mathbf{b}\mathbf{a}^\top + \mathbf{C}^\top \mathbf{X}\mathbf{a}\mathbf{b}^\top\right)^\top$	$\mathbf{C}\mathbf{X}\mathbf{b}\mathbf{a}^\top + \mathbf{C}^\top \mathbf{X}\mathbf{a}\mathbf{b}^\top$

自變數越多，誤差越小

自變數多一個

- 前面一個例子，假設研究人員只知道*Edu*

  $\hat{Y}$ = ***47.1378 + 5.2338 Edu***

  R square (R^2) equals **0.5097**

  Adjusted R square equals **0.4719**.

- 假如研究人員可以探知到額外的變數——家庭因素，並納入回歸，則可以得到以下回歸式

  (Model 2)
  $\hat{Y}$ = ***-1.539 + 6.0851 Edu + 8.9619 Fami***

  R- square (R^2) equals **0.7851**.

  Adjusted R square equals **0.7492**.

就學時長(年)	家庭因素	收入
Edu	Fami	Y
12	8	148
18	1	111
18	5	159
18	3	140
18	7	194
22	1	135
12	4	85
6	5	63
12	4	135
12	7	114
12	3	115
6	2	71
9	5	94
12	3	91
18	4	125

https://www.statskingdom.com/410multi_linear_regression.html

(續)

- 比較Model 1與Model 2

	(Error) ^2	R-Sqr	Adjusted R-Sqr
Model 1	**8,416.3**	**0.5097**	**0.4719**
Model 2 多知道了「家庭狀況」	3,689.1	**0.7850**	**0.7492**

誤差變小了
「偏誤平方和變小」

模型可解釋的資料變多

推論

Inference

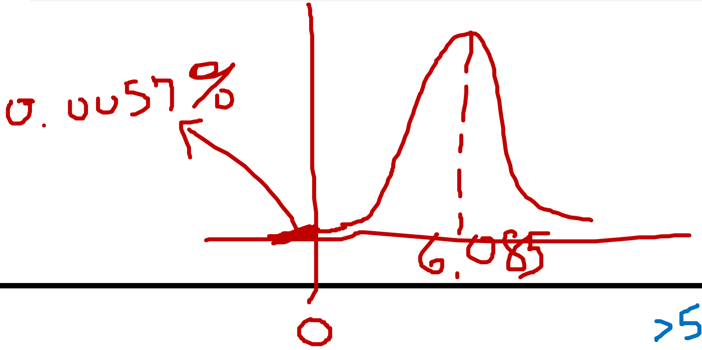
- ※動手試試看：　https://www.statskingdom.com/

預測
推論 Inference

- 什麼是t-value, P-value?

Student's t

	Coeff	SE	t-stat	lower $t_{0.025}(12)$	upper $t_{0.975}(12)$	Stand Coeff	p-value	VIF
b	-1.539015	18.825981	-0.0817495	-42.557304	39.479273	0	0.936194	
EDU	6.085108	1.004914	6.055354	3.895589	8.274627	0.830022	0.0000571173	1.048959
FAMI	8.961905	2.285552	3.921112	3.982116	13.941694	0.537476	0.00203103	1.048959



誤差是不全知

Bias is without Omniscience

(Non omniscius)

公車的英文/西方語言?

起司漢堡

如果可觀測到「福」

Model 3

就學時長(年)	家庭因素	福分	收入
Edu	Fami	Fu	Y
12	8	18	148
18	1	9	111
18	5	18	159
18	3	9	140
18	7	30	194
22	1	3	135
12	4	9	85
6	5	15	63
12	4	24	135
12	7	15	114
12	3	0	115
6	2	15	71
9	5	0	94
12	3	12	91
18	4	9	125

- 變數：
  - Edu
  - Fami
  - **Fu ← 假設觀測的到**
- 結果：

(Model 3)

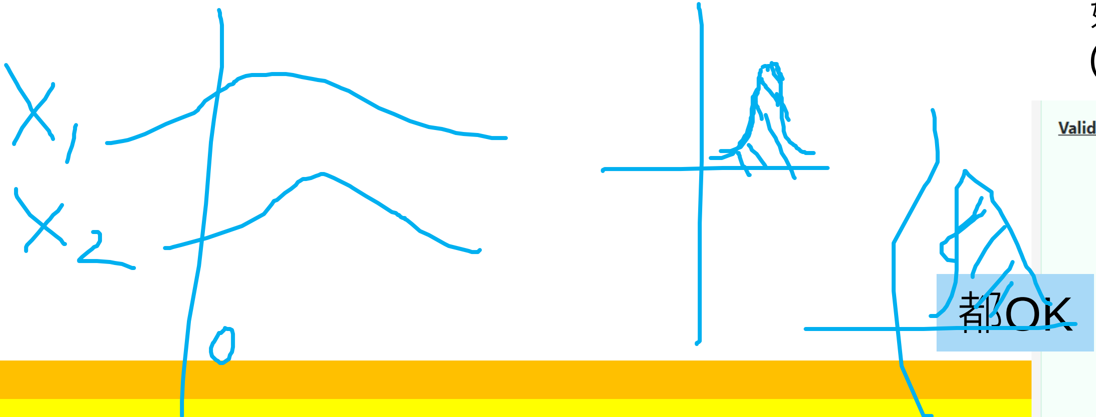
- ***$\hat{Y}$ = -2.7756 + 5.8865 Edu + 6.4951 Fami + 1.1409 Fu***
- R-Square = 0.8376
- Adjusted R square (R^2) = 0.7934

	Coeff	SE	t-stat
b	-2.7756	17.099	-0.1623
Edu	5.8865	0.9181	6.4116
Fami	6.4951	2.4512	2.6497
Fu	1.1409	0.604	1.8889

ANOVA table

Source	DF	Sum of Square	Mean Square	F Statistic	P-value
Regression (between $\hat{y}_i$ and $\bar{y}$)	3	14377.6993	4792.5664	18.925	0.0001189
Residual (between y_i and $\hat{y}_i$)	11	2785.6341	253.2395		
Total (between y_i and $\bar{y}$)	14	17163.3333	1225.9524		

	Coeff	SE	t-stat	lower $t_{0.05}(11)$	upper $t_{0.95}(11)$	Stand Coeff	p-value	VIF
b	-2.7756	17.099	-0.1623	-33.4834	27.9322	0	0.874	
Edu	5.8865	0.9181	6.4116	4.2377	7.5353	0.8029	0.00005	1.0629
Fami	6.4951	2.4512	2.6497	2.0929	10.8972	0.3895	0.0226	1.4647
Fu	1.1409	0.604	1.8889	0.05615	2.2256	0.2712	0.08555	1.3968



如果兩個變數之間有「共線性」
(collinearity)→必須拿掉其中一個變數

Validation

- **Residual normality**
  linear regression assumes normality for residual errors. Shapiro Wilk p-value equals 0.3168. It is assumed that the data is normally distributed.

- **Homoscedasticity - homogeneity of variance**
  The White test p-value equals 0.5475 (F=0.6337). It is assumed that the variance is homogeneous.

- **Multicollinearity - intercorrelations among the predictors (X_i)**
  There is no multicollinearity concern as all the VIF values are smaller than 2.5 .

(續)

- 比較Model 2與Model 3

	(Error) ^2	R-Sqr	Adjusted R-Sqr
Model 2	3,689.1	**0.7850**	**0.7492**
Model 3 多知道「福份」	**2,785.6**	**0.8376**	**0.7934**

「偏誤平方和變小」縮到更小

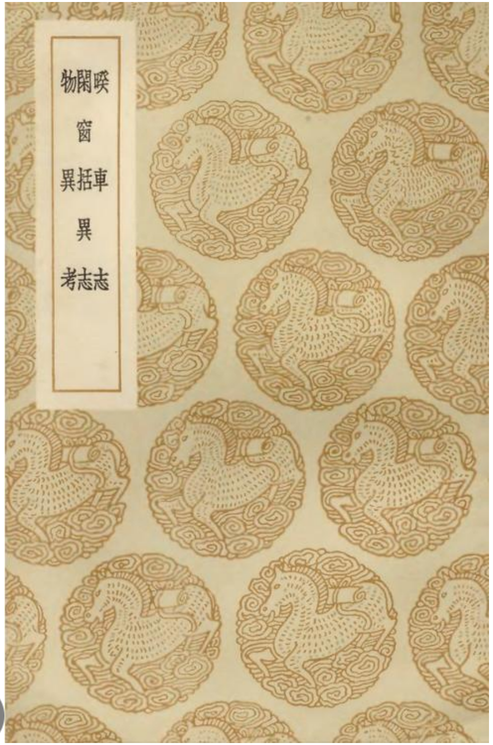
可見有許多誤差 (誤差的平方) 是因為「不全知」所造成
如果全知，應該會零誤差

禍──福的相反

雷殛案

- 發生頻率比車禍碰撞稍低
- 「… … 紹興甲寅七月十四日，吳縣光福雅宜山一村夫，以事私恨其母，遂萌梟獍之心，懷刃挈榼，與母同之近村看親。中路，請母藉草飲，意欲乘醉行逆。時天晴霽，俄有黑雲驟起，大震一聲，擊其子殞道旁。母初不知，而怪其衣中懷刃，有知其謀者，始以告焉。… …（范公懋德老承務說九事）」──《睽車志》卷二

https://zh.wikisource.org/wiki/%E7%9D%BD%E8%BB%8A%E5%BF%97_(%E5%9B%9B%E5%BA%AB%E5%85%A8%E6%9B%B8%E6%9C%AC)/%E5%8D%B72



形自言即其妾引決死矣懇求為誦經追脩士人大憂恐亟遣僕歸為其區處暨僕還得家信則妾故無恙鬼復夜至士人詰其妾欲奏章治之鬼哀祈實非妾因公憂慮之切故假此以覬薦拔自此不敢復出幸勿見治但今業已至此不能獨迴須且相隨以俟公歸許之自此悄然士人幹畢將還約親故十人同遊鍾山士人先至憩僧房以俟忽復聞鬼語士人方怒叱之乃云非敢為厲有小事奉報耳九客皆已至

山下其間第幾人乘騾第幾人騎白馬此二人他日貴人也問何以知之曰二人所過鬼物皆避道餘則不然二人者葉審言樞密其一也時方為小官云

紹興甲寅七月十四日吳縣光福雅宜山一村夫以事私恨其母遂萌梟獍之心懷刃挈榼與母同之近郵看親中路請母藉草飲意欲乘醉行逆時天晴霽俄有黑雲驟起大震一聲擊其子殞道傍母初不知而怪其衣中懷刃有知其謀者始以告焉又長洲縣北

來源：(起源於宋)《睽車志》清朝四庫全書版

原村農夫謝三二不敬其母動有悖言乾道庚寅夏
五月雨霽欲放田水詈母而出纔至田所大雷震死
范公懋德老承務說九事

紹興五年六月大雷電無錫蘇村一民家所用斗秤盡
挂於門外大樹之杪行人皆見之蓋其家每輕重其
手也

紹興三年癸丑八月五日平江長洲縣地震自西北方來
樹林皆搖動父老云元祐九年九月二十一日已嘗

如此又紹興十三年癸亥三月十五日清明大雪盈
尺

熙寧間有人授泗州盱眙令自陳乞改名雍觀時王荆
公當國怪其名無義理因問改名之故對云夢中神
告如此固亦自不曉其義後其人之官一日自城還
邑從吏卒行渡浮橋忽大風驟起鼓其衣裾盡沒淮
水已而從者拯救皆免獨不得令事聞朝廷荆公曰
向見此人無故改名且疑雍觀二字或有出處因閱

來源：(起源於宋)《睽車志》清朝四庫全書版

(其二)

- 楊○德，登船遊樂，電話響起，到外面甲板接電話
    - 原本天氣良好，突然漂來雲霧，能見度突然變低。
    - 遊艇與運煤船相撞。
    - 為了救援楊某，遊艇水手拋錨來緊急停止
    - 錨恰好擊中楊某頭部，楊某身亡

祖上積德

- 餘杭陳醫，有貧人病危，陳治之痊，亦不責報。後陳因避雨過其家，其姑令婦伴宿以報恩，婦唯唯，夜深就之曰：「君救妾夫，此姑意也。」陳見婦少而美，亦心動，隨力制之，自語曰：「不可！」婦強之，陳連曰：「不可不可！」坐以待旦，最後幾不自持，又大呼曰：「不可二字最難！」天明遁去。陳有子應試，主試棄其文，忽聞呼曰：「不可！」挑燈複閱，再棄之，又聞連聲呼曰：「不可不可！」最後決意棄之，忽聞大呼曰：「不可二字最難！」連聲不已，因錄之。榜後召問故，其子亦不解，歸告父。父曰：「此我壯年事也，不意天之報我如此！」

—《不可錄》

雙元變數

- 雙元變數(binary variables)
  - 又稱二元變數、啞變數(dummy variables)、啞(dummy)
- 例如，如果張叅是男生，此變數為1、 李姒是女生，此變數為0

就學時長(年)	家庭因素	男	收入
Edu	Fami	Male	Y
12	8	1	148
18	1	1	111
18	5	1	159
18	3	1	140
18	7	1	194
22	1	1	135
12	4	0	85
6	5	0	63
12	4	0	135
12	7	1	114
12	3	1	115
6	2	1	71
9	5	1	94
12	3	0	91
18	4	1	125

(Model 4)

$$\hat{Y} = -1.7838 + 5.7494\ Edu + 8.7287\ Fami + 7.9041\ Male$$

R square (R^2) = **0.7937**.

Adjusted R square equals **0.7375**.

Correlation matrix (pearson)

	Y	Edu	Fami	Male
Y	1	0.7139	0.3582	0.4486
Edu	0.7139	1	-0.216	0.4139
Fami	0.3582	-0.216	1	0.03963
Male	0.4486	0.4139	0.03963	1

```
if(!"car" %in% installed.packages()){install.packages("
car")}
library("car")
y<-c(148,111,159,140,194,135,85,63,135,114,115,71,94,91,
125)
edu<-c(12,18,18,18,18,22,12,6,12,12,12,6,9,12,18)
fami<-c(8,1,5,3,7,1,4,5,4,7,3,2,5,3,4)
male<-c(1,1,1,1,1,1,0,0,0,1,1,1,1,0,1)
model4 = lm(y~edu+fami+male)
```

ANOVA table

Source	DF	Sum of Square	Mean Square	F Statistic	P-value
Regression (between $\hat{y}_i$ and $\bar{y}$)	3	13622.8663	4540.9554	14.1085	0.0004349
Residual (between y_i and $\hat{y}_i$)	11	3540.467	321.8606		
Total (between y_i and $\bar{y}$)	14	17163.3333	1225.9524		

	Coeff	SE	t-stat	lower $t_{0.05}(11)$	upper $t_{0.95}(11)$	Stand Coeff	p-value	VIF
b	-1.7838	19.2662	-0.09259	-36.3836	32.8161	0	0.9279	
Edu	5.7494	1.1407	5.0402	3.7009	7.798	0.7842	0.0003778	1.291
Fami	8.7287	2.3636	3.6929	4.4839	12.9735	0.5235	0.003545	1.0715
Male	7.9041	11.6299	0.6796	-12.9819	28.79	0.1033	0.5108	1.2327

>5% 不顯著

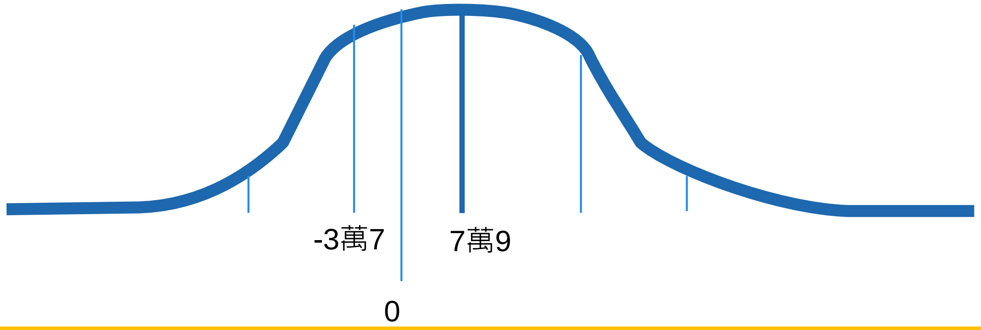
現代計量學的議題

Edu $\longrightarrow$ Y

$\Sigma \beta X$

- Male變數不顯著，是否要剔除?
  - 端看有無成為「控制變數」的要件
  - 就是，你是否相信知道個體是男是女可以提升模型的準確性

  - 男生 平均可多 7 萬 9的年薪，在所有條件相同的狀況下



公式推導：

- $\mathrm{Y}|_{z=1} - \mathrm{Y}|_{z=0}$, *Ceteris paribus*

- $(\mathrm{Y}|_{z=1} / \mathrm{Y}|_{z=0})$, *Ceteris paribus*

	(Error) ^2	R-Sqr	Adjusted R-Sqr
Model 2 Edu、Fami	3,689.1	**0.7850**	**0.7492**
Model 4 Edu、Fami、Male	3,540.5	**0.7937**	**0.7375**

「偏誤和」一定會變小、R-sqr一定會變大
但是也有可能過度配適(overfit)

所以重點是合不合理?

性質：0,1訂反不影響推論結果

- 若雙元改為以Female為1:

Edu	Fami	FEMALE	Y
12	8	0	148
18	1	0	111
18	5	0	159
18	3	0	140
18	7	0	194
22	1	0	135
12	4	1	85
6	5	1	63
12	4	1	135
12	7	0	114
12	3	0	115
6	2	0	71
9	5	0	94
12	3	1	91
18	4	0	125

(Model 4b)

$$\hat{Y} = 6.1203 + 5.7494\ Edu + 8.7287\ Fami - 7.9041\ FEMALE$$

(續)

- 推論不變(常數項有變化→但會使Y-hat相同)

(Model 4)

$$\hat{Y} = -1.7838 + 5.7494\,Edu + 8.7287\,Fami + 7.9041\,Male$$

(Model 4b)

$$\hat{Y} = 6.1203 + 5.7494\,Edu + 8.7287\,Fami - 7.9041\,FEMALE$$

「控制變數」初探

A Primer on the “Control Variable”

世代(Cohort)

就學時長(年)	家庭因素	1985出生	1986出生	男	福分	收入
Edu	Fami	Coh1985	Coh1986	Male	Fu	Y
12	8	0	0	1	18	148
18	1	1	0	1	9	111
18	5	0	0	1	18	159
18	3	1	0	1	9	140
18	7	0	1	1	30	194
22	1	0	1	1	3	135
12	4	0	0	0	9	85
6	5	1	0	0	15	63
12	4	0	0	0	24	135
12	7	0	1	1	15	114
12	3	1	0	1	0	115
6	2	0	1	1	15	71
9	5	1	0	1	0	94
12	3	0	0	0	12	91
18	4	0	0	1	9	125

- 加入世代(Cohort):
  - 例如，三個世代的人
    - 1985出生 Coh_{1985} = 1; otherwise 0
    - 1986出生 Coh_{1986} = 1; otherwise 0
    - 1987出生 Coh_{1985} = 0; Coh_{1986} = 0
  - 兩個變數就呈飽和模型(saturated model)→ 因為第三個世代就是其他世代變數為0。

(Model 5)

$$\hat{Y} = 1.0487 + 5.6464\ Edu + 8.4829\ Fami - 3.0718\ Coh_{1985} + 1.2304\ Coh_{1986} + 8.2952\ Male$$

(續)

- 加入世代變數後，看來「世代變數」都不顯著：

	Coeff	SE	t-stat	lower $t_{0.025}(9)$	upper $t_{0.975}(9)$	Stand Coeff	p-value	VIF
b	1.0487	25.2526	0.04153	-56.0766	58.174	0	0.9678	
Edu	5.6464	1.3678	4.1282	2.5524	8.7405	0.7702	0.002566	1.5364
Fami	8.4829	2.8276	3.0001	2.0865	14.8793	0.5088	0.01495	1.2694
Coh1985	-3.0718	14.3034	-0.2148	-35.4284	29.2847	-0.04281	0.8347	1.7539
Coh1986	1.2304	14.9148	0.0825	-32.5092	34.97	0.01609	0.9361	1.6782
Male	8.2952	15.4191	0.538	-26.5852	43.1755	0.1084	0.6036	1.7936

- 「**純統計**」老觀點：刪掉不顯著的變數
- 「**計量(econometrics)**」觀點：控制變數要保留

- 事實上，誰比較準?

(Model 4)

$\hat{Y}$ = -1.7838 + 5.7494 Edu + 8.7287 Fami + 7.9041 Male

(Model 5)

$\hat{Y}$ = 1.0487 + 5.6464 Edu + 8.4829 Fami - 3.0718 Coh_{1985} + 1.2304 Coh_{1986} + 8.2952 Male

- … 以上各欄數據由以下產生

(真實答案)

Y = 5 Edu + 4 Fami -2 Coh1985 +1 Coh1986 + 10 Male + Fu + De

- ※一般實證研究，不會知道正確答案，這裡是為了說明，把福、德藏起來當未知數

(人類不可觀測)→進行回歸 看到底回歸式是否真的準或不準。

- 1985、1982年出生較坎坷→建模者(modeler)沒辦法定變數說學測遇到地震颱風 → cohort變數涵蓋

(真實答案)

Y = 5 Edu + 4 Fami -2 Coh1985 +1 Coh1986 + 10 Male + Fu + De

(真實答案)

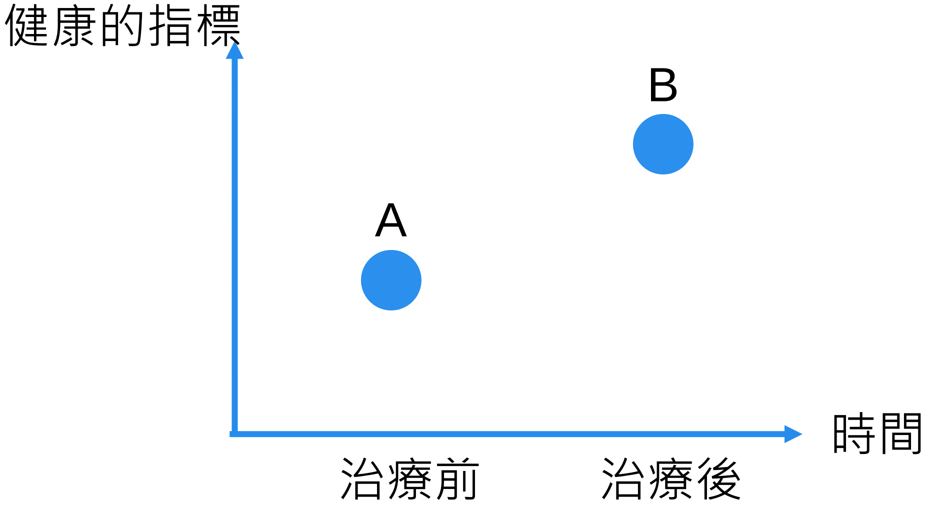
Y = 5 Edu + 4 Fami -2 **(是否國三遇到颱風地震，影響讀書)** + 1 **(母親生產受到1987韋恩颱風影響，)** + 10 Male + 1福 + 1德

時間趨勢

藥物實驗的案例

A Drug Trial

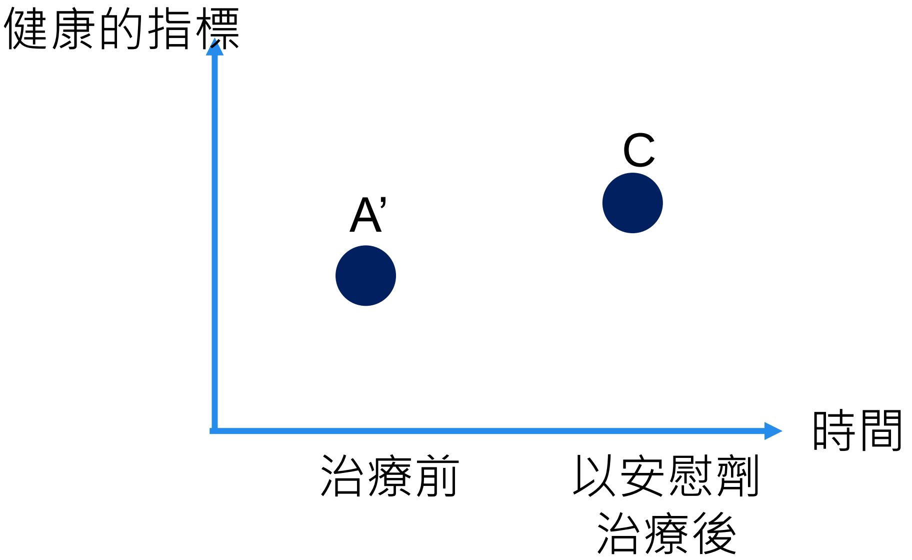
- 服用某種藥物後，症狀減輕：
  - 藥物的效果是(B – A) 嗎?



安慰劑效應

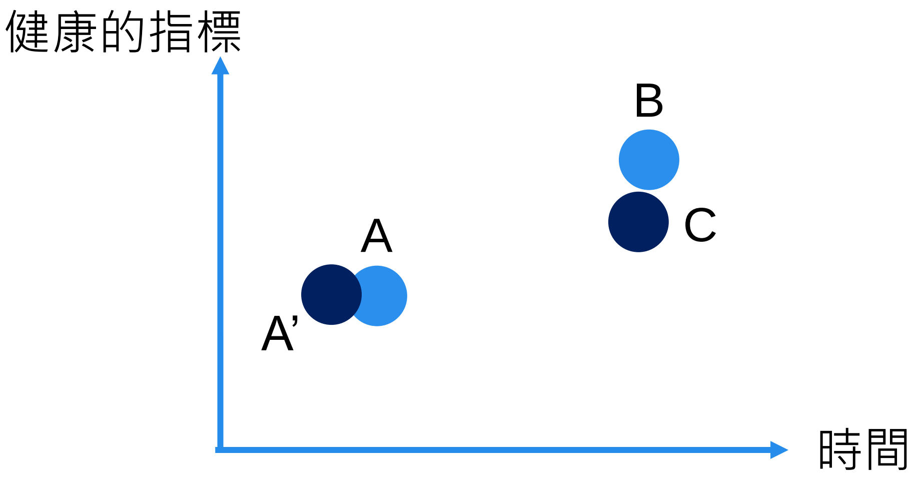
Placebo Effect

- 安慰劑——由糖水等其他非藥物製成
- 但告訴患者藥物治療：



對策的效用

- 考慮安慰劑效應後，藥物真正的效用為{ (B – A) – (C – A’) }



嚴謹的數學式：

建模

- 「Difference-in-difference (D-i-D)」
  - Effect $E = [\,(\bar{y}_{\text{after}} \mid D = 1) - (\bar{y}_{\text{before}} \mid D = 1)] - [(\bar{y}_{\text{after}} \mid D = 0) - (\bar{y}_{\text{before}} \mid D = 0]$

$\bar{y}_{\text{after}}$:事後的平均健康指標

$\bar{y}_{\text{before}}$:事前的平均健康指標

$D = 1$:患者服用欲研究的藥物

$D = 0$:患者服用欲研究的藥物

  - 模型： $y_{i,t} = \beta_1 + \beta_2 T_t + \beta_3 D_i + \delta(\boldsymbol{T_t} \times \boldsymbol{D_i}) + \varepsilon_{i,t}$

T_t= 1 事後(雙元)

D_i= 1 觀測組(雙元)

$\beta_1, \beta_2, \beta_3, \delta$ 欲求得的參數

Greene, p.168

(續)

$$y_{i,t} = \beta_1 + \beta_2 T_t + \beta_3 D_i + \delta(\boxed{\boldsymbol{T_t \times D_i}}) + \varepsilon_{i,t}$$

積項(cross product term)

- 為何需要交叉項? 協助估計真正的效用。

(1) $(y_{i,t=2}|D_i = 1) - (y_{i,t=1}|D_i = 1)$ =

(2) $(y_{i,t=2}|D_i = 0) - (y_{i,t=1}|D_i = 0)$ =

(1)-(2) =

D-i-D案例

•

2022.4.16 SATURDAY

6.3 DIFFERENCE IN DIFFERENCES REGRESSION

Many recent studies have examined the causal effect of a $\boxed{\text{treatment}}$ on some kind of **response**. Examples include the effect of attending an elite college on lifetime income [Dale and Krueger (2002, 2011)], the effect of cash transfers on child health [Gertler (2004)], the effect of participation in job training programs on income [LaLonde (1986)], the effect on employment of an increase in the minimum wage in one of two neighboring states [Card and Krueger (1994)] and pre- versus post-regime shifts in macroeconomic models [Mankiw (2006)], to name but a few.

6.3.1 TREATMENT EFFECTS

The applications can often be formulated in regression models involving a treatment dummy variable, as in

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \delta D_i + \varepsilon_i,$$

where the shift parameter, δ (under the right assumptions), measures the causal effect of the treatment or the policy change (conditioned on $\mathbf{x}$) on the sampled individuals. For example, Table 6.6 provides a log wage equation based on a national (U.S.) panel survey. One of the variables is *UNION*, a dummy variable that indicates union membership. Measuring the effect of union membership on wages is a longstanding objective in labor economics—see, for example, Card (2001). Our estimate in Table 6.6 is roughly 0.08, or 8%. It will take a bit of additional specification analysis to conclude that the *UNION* dummy truly does measure the effect of membership in that context.[10]

In the simplest case of a comparison of one group to another, without covariates,

$$y_i = \beta_1 + \delta D_i + \varepsilon_i.$$

Least squares regression of y on D will produce

$$b_1 = (\bar{y} \mid D_i = 0),$$

that is, the average outcome of those who did not experience the treatment, and

$$d = (\bar{y} \mid D_i = 1) - (\bar{y} \mid D_i = 0),$$

the difference in the means of the two groups. Continuing our earlier example, if we measure the *UNION* effect in Table 6.6 without the covariates, we find

$$\ln Wage = 6.673\ (0.023) + 0.00834\ UNION\ (0.028).$$

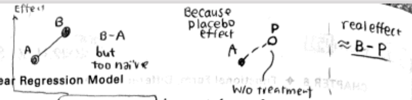
(Standard errors are in parentheses.) Based on a simple comparison of means, there appears to be a less than 1% impact of union membership. This is in sharp contrast to the 8% reported earlier.

When the analysis is of an intervention that occurs over time to everyone in the sample, such as in Krueger's (1999) analysis of the Tennessee STAR experiment in which school performance measures were observed before and after a policy that dictated a change in class sizes, the treatment dummy variable will be a period indicator, $T_t = 0$ in period 1 and 1 in period 2. The effect in β_2 then measures the change in the outcome variable, for example, school performance, pre- to post-intervention; $b_2 = \bar{y}_1 - \bar{y}_0$.

[10] See, for example, Angrist and Pischke (2009, pp. 221–225.)

Source: Greene (2018)

Naïve before & after



The assumption that the $\boxed{\text{treatment group}}$ does not change from period 1 to period 2 (or that the treatment group and the control group look the same in all other respects) weakens this analysis. A strategy for strengthening the result is to include in the sample a group of $\boxed{\textbf{control observations}}$ that do not receive the treatment. The change in the outcome for the **treatment group** can then be compared to the change for the **control group** under the presumption that the difference is due to the intervention. An intriguing application of this strategy is often used in clinical trials for health interventions to accommodate the **placebo effect**. The placebo effect is a controversial, but apparently tangible outcome in some clinical trials in which subjects "respond" to the treatment even when the treatment is a decoy intervention, such as a sugar or starch pill in a drug trial.[11] A broad template for assessment of the results of such a clinical trial is as follows: The subjects who receive the placebo are the controls. The outcome variable—level of cholesterol, for example—is measured at the baseline for both groups. The treatment group receives the drug, the control group receives the placebo, and the outcome variable is measured pre- and post-treatment. The impact is measured by the difference in differences, (D-I-D)

$$E = [(\bar{y}_{exit} \mid treatment) - (\bar{y}_{baseline} \mid treatment)] - [(\bar{y}_{exit} \mid placebo) - (\bar{y}_{baseline} \mid placebo)].$$

The presumption is that the difference in differences measurement is robust to the placebo effect *if it exists*. If there is no placebo effect, the result is even stronger (assuming there is a result).

A common social science application of treatment effect models is in the evaluation of the effects of discrete changes in policy.[12] A pioneering application is the study of the Manpower Development and Training Act (MDTA) by Ashenfelter and Card (1985) and Card and Krueger (2000). A widely discussed application is Card and Krueger's (1994) analysis of an increase in the minimum wage in $\boxed{\text{New Jersey.}}$ The simplest form of the model is one with a pre- and post-treatment observation on a group, where the outcome variable is y, with

$$y_{it} = \beta_1 + \beta_2 \underbrace{T_t}_{\text{period}} + \beta_3 \underbrace{\boxed{D_i}}_{\text{received or not}} + \delta(T_t \times D_i) + \varepsilon_{it}, \quad t = 0, 1. \qquad (6\text{-}3)$$

*** KEY Formulations in D-I-D

the PRODUCT of the dummies

In this model, T_t is a dummy variable that is zero in the pre-treatment period and one after the treatment and D_i equals one for those individuals who received the treatment. The change in the outcome variable for the treated individuals will be

$$(y_{i2} \mid D_i = 1) - (y_{i1} \mid D_i = 1) = (\beta_1 + \beta_2 + \beta_3 + \delta) - (\beta_1 + \beta_3) = \beta_2 + \delta.$$

For the controls, this is

$$(y_{i2} \mid D_i = 0) - (y_{i1} \mid D_i = 0) = (\beta_1 + \beta_2) - \beta_1 = \beta_2.$$

The difference in differences is (D-I-D)

$$[(y_{i2} \mid D_i = 1) - (y_{i1} \mid D_i = 1)] - [(y_{i2} \mid D_i = 0) - (y_{i1} \mid D_i = 0)] = \delta.$$

[11] See Hróbjartsson and Götzsche (2001).

[12] Surveys of literatures on treatment effects, including use of "D-i-D" estimators, are provided by Imbens and Wooldridge (2009), Millimet, Smith, and Vytlacil (2008), Angrist and Pischke (2009), and Lechner (2011).

Source: Greene (2018)

DID $(B-A)-(P-A')$

B

P

placebo

In the multiple regression of y_{it} on a constant, T, D, and $T \times D$, the least squares estimate of δ will equal the difference in the changes in the means,

$$
\begin{aligned}
d &= (\bar{y}|D = 1, \textit{Period } 2) - (\bar{y}|D = 1, \textit{Period } 1) \\
&\quad - (\bar{y}|D = 0, \textit{Period } 2) - (\bar{y}|D = 0, \textit{Period } 1) \\
&= \Delta\bar{y}|\textit{treatment} - \Delta\bar{y}|\textit{control}.
\end{aligned}
$$

The regression is called a difference in differences estimator in reference to this result.

Example 6.8 SAT Scores

Each year, about 1.7 million American high school students take the SAT test. Students who are not satisfied with their performance have the opportunity to retake the test. Some students take an SAT prep course, such as Kaplan or Princeton Review, before the second attempt in the hope that it will help them increase their scores. An econometric investigation might consider whether these courses are effective in increasing scores. The investigation might examine a sample of students who take the SAT test twice, with scores y_{i0} and y_{i1}. The time dummy variable T_t takes value $T_0 = 0$ "before" and $T_1 = 1$ "after." The treatment dummy variable is $D_i = 1$ for those students who take the prep course and 0 for those who do not. The applicable model would be (6-3),

KEY FORMULATIONS in D-T-D

$$\textit{SAT Score}_{i,t} = \beta_1 + \beta_2\, \textit{2ndTest}_t + \beta_3\, \textit{PrepCourse}_i + \delta\, \textit{2ndTest}_t \times \textit{PrepCourse}_i + \varepsilon_{i,t}.$$

The estimate of δ would, in principle, be the treatment, or prep course effect.

(not main topic here) they will be confounded by the motivations in ε_{it}

This small example illustrates some major complications. First, and probably most important, the setting does not describe a randomized experiment such as the clinical trial suggested earlier would be. The treatment variable, *PrepCourse*, would naturally be taken by those who are persuaded that it would provide a benefit—that is, the treatment variable is not an exogenous variable. Unobserved factors that are likely to contribute to higher test scores (and are embedded in $\boxed{\varepsilon_{i,t}}$) would likely motivate the student to take the prep course as well. This *selection effect* is a compelling confounder of studies of treatment effects when the treatment is voluntary and self selected. Dale and Krueger's (2002, 2011) analysis of the effect of attendance at an elite college provides a detailed analysis of this issue. Second, test performance, like other performance measures, is probably subject to regression to the mean—there is a negative autocorrelation in such measures. In this regression context, an unusually high disturbance in period 0, all else equal, would likely be followed by a low value in period 1. Of course, those who achieve an unusually high test score in period 0 are less likely to return for the second attempt. Together with the selection effect, this produces a very muddled relationship between the outcome and the test preparation that is estimated by least squares. Finally, it is possible that there are other measurable factors (covariates) that might contribute to the test outcome or changes in the outcome. A more complete model might include these covariates. We do note any such variable $x_{i,t}$ would have to vary between the first and second test, else they would simply be absorbed in the constant term.

R-T-M

When the treatment is the result of a policy change or event that occurs completely outside the context of the study, the analysis is often termed a natural experiment. Card's (1990) study of a major immigration into Miami in 1979 is an application.

Example 6.9 A Natural Experiment: The Mariel Boatlift

A sharp change in policy can constitute a natural experiment. An example studied by Card (1990) is the Mariel boatlift from Cuba to Miami (May–September 1980), which increased the Miami labor force by 7%. The author examined the impact of this abrupt change in labor market conditions on wages and employment for nonimmigrants. The model compared Miami (the treatment group) to a similar city, Los Angeles (the control group). Let i denote an

Source: Greene

individual and D denote the "treatment," which for an individual would be equivalent to "lived in the city that experienced the immigration." For an individual in either Miami or Los Angeles, the outcome variable is

$$Y_i = 1 \quad \text{if they are unemployed and 0 if they are employed.}$$

Let c denote the city and let t denote the period, before (1979) or after (1981) the immigration. Then, the unemployment rate in city c at time t is $E[y_{i,0}|c, t]$ if there is no immigration and it is $E[y_{i,1}|c, t]$ if there is the immigration. These rates are assumed to be constants. Then

$$
\begin{aligned}
E[y_{i,0}|c, t] &= \beta_t + \gamma_c \quad \text{without the immigration,} \\
E[y_{i,1}|c, t] &= \beta_t + \gamma_c + \delta \quad \text{with the immigration.}
\end{aligned}
$$

The effect of the immigration on the unemployment rate is measured by δ. The natural experiment is that the immigration occurs in Miami and not in Los Angeles but is not a result of any action by the people in either city. Then,

$$
\begin{aligned}
&E[y_i|M, 79] = \beta_{79} + \gamma_M \quad \textit{and} \quad E[y_i|M, 81] = \beta_{81} + \gamma_M + \delta \quad \textit{for Miami,} \\
&E[y_i|L, 79] = \beta_{79} + \gamma_L \quad \textit{and} \quad E[y_i|L, 81] = \beta_{81} + \gamma_L \quad \textit{for Los Angeles.}
\end{aligned}
$$

It is assumed that unemployment growth in the two cities would be the same if there were no immigration. If neither city experienced the immigration, the change in the unemployment rate would be

$$
\begin{aligned}
E[y_{i,0}|M, 81] - E[y_{i,0}|M, 79] &= \beta_{81} - \beta_{79} \quad \textit{for Miami,} \\
E[y_{i,0}|L, 81] - E[y_{i,0}|L, 79] &= \beta_{81} - \beta_{79} \quad \textit{for Los Angeles.}
\end{aligned}
$$

If both cities were exposed to migration,

$$
\begin{aligned}
E[y_{i,1}|M, 81] - E[y_{i,1}|M, 79] &= \beta_{81} - \beta_{79} + \delta \quad \textit{for Miami,} \\
E[y_{i,1}|L, 81] - E[y_{i,1}|L, 79] &= \beta_{81} - \beta_{79} + \delta \quad \textit{for Los Angeles.}
\end{aligned}
$$

Only Miami experienced the immigration (the "treatment"). The difference in differences that quantifies the result of the experiment is

$$\{E[y_{i,1}|M, 81] - E[y_{i,1}|M, 79]\} - \{E[y_{i,0}|L, 81] - E[y_{i,0}|L, 79]\} = \delta.$$

The author examined changes in employment rates and wages in the two cities over several years after the boatlift. The effects were surprisingly modest (essentially nil) given the scale of the experiment in Miami.

Example 6.10 Effect of the Minimum Wage

Card and Krueger's (1994) widely cited analysis of the impact of a change in the minimum wage is similar to Card's analysis of the Mariel Boatlift. In April 1992, New Jersey (NJ) raised its minimum wage from $\boxed{\$4.25 \text{ to } \$5.05.}$ The minimum wage in neighboring Pennsylvania (PA) was unchanged. The authors sought to assess the impact of this policy change by examining the change in employment in the two states from February to November, 1992 at fast food restaurants that tended to employ large numbers of people at the minimum wage. Conventional wisdom would suggest that, all else equal, whatever labor market trends were at work in the two states, NJ's would be affected negatively by the abrupt 19% wage increase for minimum wage workers. This certainly qualifies as a natural experiment. NJ restaurants could not opt out of the treatment. The authors were able to obtain data on employment for 331 NJ restaurants and 97 PA restaurants in the first wave. Most of the first wave restaurants provided data for the second wave, 321 and 78, respectively. One possible source of "selection" would be attrition from the sample. Though the numbers are small, the possibility that the second wave sample was substantively composed of firms that were affected by the policy change

Source: Greene

TABLE 6.8 Full Time Employment in NJ and PA Restaurants

	PA	*NJ*
First Wave (February)	23.33	20.44
Second Wave (November)	21.17	21.03
Difference	−2.16	0.59
Difference (balanced)	−2.28	0.47

would taint the analysis (e.g., if firms were driven out of business because of the increased labor costs). The authors document at some length the data collection process for the second wave. Results for their experiment are shown in Table 6.8.

The first reported difference uses the full sample of available data. The second uses the "balanced sample" of all stores that reported data in both waves. In both cases, the difference in differences would be

$$\Delta(\text{NJ}) - \Delta(\text{PA}) = +2.75 \text{ full time employees.}$$

A superficial analysis of these results suggests that they go in the wrong direction. Employment rose in NJ compared to PA in spite of the increase in the wage. Employment would have been changing in both places due to other economic conditions. The policy effect here might have distorted that trend. But, it is also possible that the trend in the two states was different. It has been assumed throughout so far that it is the same. Card and Krueger (2000) examined this possibility in a followup study. The newer data cast some doubt on the crucial assumption that the trends were the same in the two states.

Card and Krueger (1994) considered the possibility that restaurant specific factors might have influenced their measured outcomes. The implied regression would be

$$y_{it} = \beta_2 \boxed{T_t} + \beta_3 \boxed{D_i} + \delta \boxed{T_t \times D_i} + (\alpha_i + \gamma' \mathbf{x}_i) + \varepsilon_{it}, \quad t = 0, 1.$$

Note the individual specific constant term that represents the unobserved heterogeneity and the addition to the regression. In the restaurant study, $\mathbf{x}_i$ was characteristics of the store such as chain store type, ownership, and region—all features that would be the same in both waves. These would be fixed effects. In the difference in differences context, while they might indeed be influential in the outcome levels, it is clear that they will fall out of the differences:

$$\Delta E[y_{it} | D_{it} = 0, \mathbf{x}_i] = \beta_2 \qquad + \Delta(\alpha_i + \gamma' \mathbf{x}_i),$$
$$\Delta E[y_{it} | D_{it} = 1, \mathbf{x}_i] = \beta_2 + \delta + \Delta(\alpha_i + \gamma' \mathbf{x}_i).$$

The final term in both cases is zero, which leaves, as before,

$$\Delta E[y_{it} | D_{it} = 1, \mathbf{x}_i] - \Delta E[y_{it} | D_{it} = 1, \mathbf{x}_i] = \delta.$$

The useful conclusion is that in analyzing differences in differences, time invariant characteristics of the individuals will not affect the conclusions.

The analysis is more complicated if the control variables, $\mathbf{x}_{it}$, do change over time. Then,

$$y_{it} = \beta_2 T_t + \beta_3 D_i + \delta T_t \times D_i + \gamma' \mathbf{x}_{it} + \varepsilon_{it}, \quad t = 0, 1.$$

2022.4.16 SAT

Then,

$$\Delta E[y_{it} | \mathbf{x}_{it}, D_{it} = 1] = \beta_2 + \delta + \gamma'[\Delta \mathbf{x}_{it} | D_{it} = 1]$$
$$\Delta E[y_{it} | \mathbf{x}_{it}, D_{it} = 0] = \beta_2 + \qquad \gamma'[\Delta \mathbf{x}_{it} | D_{it} = 0]$$

$$\Delta E[y_{it} | D_{it} = 1, \mathbf{x}_i] - \Delta E[y_{it} | D_{it} = 1, \mathbf{x}_i] = \delta + \gamma'[(\Delta \mathbf{x}_{it} | D_{it} = 1) - (\Delta \mathbf{x}_{it} | D_{it} = 0)].$$

Now, if the effect of D_{it} is measured by the simple difference of means, the result will consist of the causal effect plus an additional term explained by the difference of the changes in the control variables. If individuals have been carefully sampled so that treatment and controls look the same in both periods, then the second effect might be ignorable. If not, then the second part of the regression should become part of the analysis.

2019 Mar 14 THUR

6.3.2 EXAMINING THE EFFECTS OF DISCRETE POLICY CHANGES

The differences in differences result provides a convenient methodology for studying the effects of exogenously imposed policy changes. We consider an application from a recent antitrust case.

Example 6.11 Difference in Differences Analysis of a Price Fixing Conspiracy[13]

Roughly 6.5% of all British schoolchildren, and more than 18% of those over 16, attend 2,600 independent fee-paying schools. Of these, roughly 10.5% are "boarders"—the remainder attend on a day basis. Each year from 1997 until June, 2003, a group of 50 of these schools shared information about intended fee increases for boarding and day students. The information was exchanged via a survey known as the "Sevenoaks Survey" (SS). The UK Office of Fair Trading (OFT, Davies (2012)) determined that the conspiracy, which was found to lead to higher fees, was prohibited under the antitrust law, the Competition Act of 1998. The OFT intervention consisted of a modest fine (10,000GBP) on each school, a mandate for the cartel to contribute about 3,000,000GBP to a trust, and prohibition of the Sevenoaks Survey. The OFT investigation was ended in 2006, but for the purposes of the analysis, the intervention is taken to have begun with the 2004/2005 academic year.

The authors of this study investigated the impact of the OFT intervention on the boarding and day fees of the Sevenoaks schools using a difference in differences regression. The pre-intervention period is academic years 2001/02 to 2003/04. The post-intervention period extends to 2011/2012. The sample consisted of the treatment group, the 50 Sevenoaks schools, and 178 schools that were not party to the conspiracy and therefore, not impacted by the treatment. (Not necessarily. More on that below.) The "balanced panel data set" of 12 years times 228 schools, or 2,736 observations, was reduced by missing data to 1,829 for the day fees model and 1,317 for the boarding fees model. Figure 6.2 (Figures 2 and 3 from the study) shows the behavior of the boarding and day fees for the schools for the period of the study.[14] It is difficult to see a difference in the rates of change of the fees. The difference in the levels is obvious, but not yet explained.

A difference in differences methodology was used to analyze the behavior of the fees. Two key assumptions are noted at the outset.

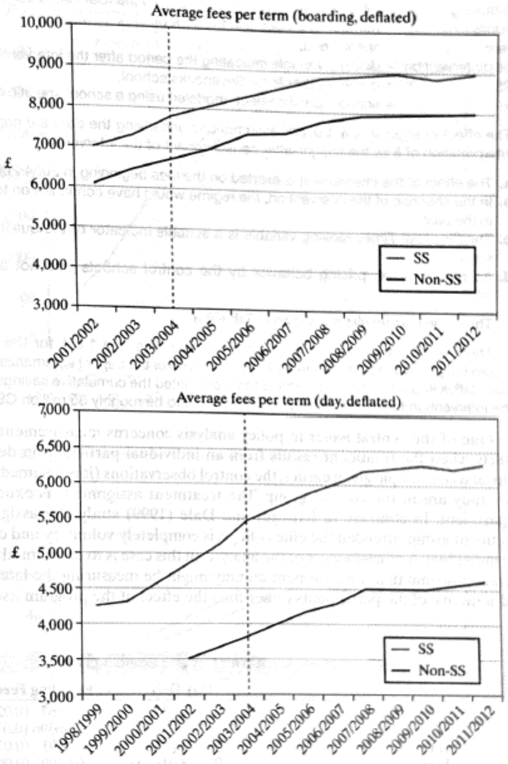
1. The schools in the control group are not affected by the intervention. This may not be the case. The non-SS schools compete with the SS schools on a price basis. If the pricing behavior of the SS schools is affected by the intervention, that of the non-SS schools may be as well.

[13] This case study is based on UK OFT (2012), Davies (2012) and Pesarisi et al. (2015).

[14] The figures are extracted from the UK OFT (2012) working paper version of the study.

Source: Greene

Figure 6.2 Price Increases by Boarding Schools.



2. It must be assumed that the trends and influences that affect the two groups of schools outside the effect of the intervention are the same. (Recall this was an issue in Card and Krueger's analysis of the minimum wage in Example 6.10.)

The linear regression model used to study the behavior of the fees is

$$\ln Fee_{it} = \alpha_i + \beta_1 \%boarder_{it} + \beta_2 \%ranking_{it} + \beta_3 \ln pupils_{it} + \beta_4 year_t + \lambda\, postintervention_t + \delta\, SS_{it} \times postintervention_t + \varepsilon_{it}$$

Fee_{it} = inflation-adjusted day or boarding fees,
%boarder = percentage of the students who are boarders at school i in year t,
%ranking = percentile ranking of the school in *Financial Times* school rankings,
pupils = number of students in the school,
year = linear trend,
postintervention = dummy variable indicating the period after the intervention,
SS = dummy variable for Sevenoaks school,
α_i = school-specific effect, modeled using a school specific dummy variable.

The effect of interest is δ. Several assumptions underlying the data are noted to justify the interpretation of δ as the sought-after causal impact of the intervention.

a. The effect of the intervention is exerted on the fees beginning in 2004/2005.
b. In the absence of the intervention, the regime would have continued on to 2012 as it had in the past.
c. The *Financial Times* ranking variable is a suitable indicator of the quality of the ranked school.
d. As noted earlier, pricing behavior by the control schools was not affected by the intervention.

The regression results are shown in Table 6.9.

The main finding is a decline of 1.5% for day fees and 1.6% for the boarding fees. Figure 6.3 [extracted from the UK OFT (2012) version of the paper] summarizes the estimated cumulative impact of the study. The authors estimated the cumulative savings attributable to the intervention based on the results in Figure 6.3 to be roughly 85 million GBP.

One of the central issues in policy analysis concerns measurement of treatment effects when the treatment results from an individual participation decision. In the clinical trial example given earlier, the control observations (it is assumed) do not know they they are in the control group. The treatment assignment is exogenous to the experiment. In contrast, in Krueger and Dale (1999) study, the assignment to the treatment group, attended the elite college, is completely voluntary and determined by the individual. A crucial aspect of the analysis in this case is to accommodate the almost certain outcome that the treatment dummy might be measuring the latent motivation and initiative of the participants rather than the effect of the program itself. That is the

TABLE 6.9 Estimated Models for Day and Boarding Fees*

	Day Fees	Boarding Feees
% Boarder	0.7730 (0.051)**	0.0367 (0.029)
% Ranking	−0.0147 (0.019)	0.00396 (0.015)
ln Pupils	0.0247 (0.033)	0.0291 (0.021)
Year	0.0698 (0.004)	0.0709 (0.004)
Post-intervention	0.0750 (0.027)	0.0674 (0.022)
Post-intervention and SS	−0.0149 (0.007)	−0.0162 (0.005)
N	1,825	1,311
R^2	0.949	0.957

Source: Pesaresi et al. (2015), Table 1.
*Model fit by least squares. Estimated individual fixed effects not shown.
**Robust standard errors that account for possible heteroscedasticity and autocorrelation in parentheses.

Source: Greene

有些科學家的態度有點問題

- 關於安慰劑效應，錯誤的態度：
  - 要「排除安慰劑效應」才能看出藥效→繼續研發藥物
  - 誤把「安慰劑效應」當干擾

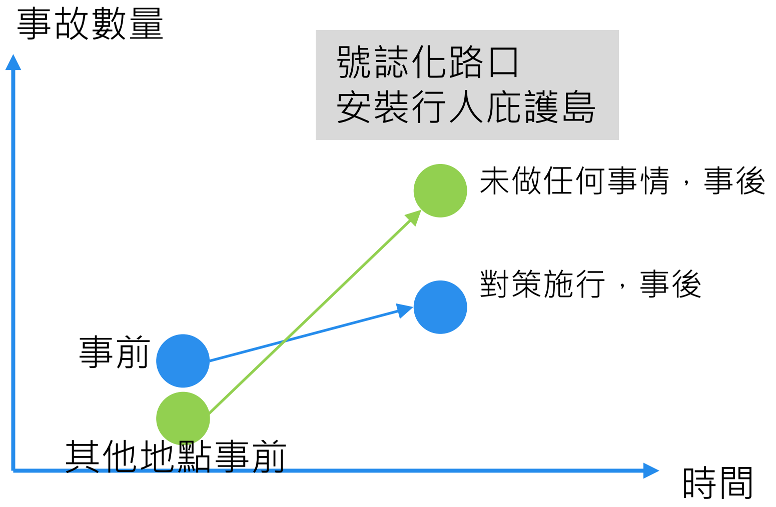
- 何不想想
  - **只靠精神上的想像 就能治癒** 背後的意義

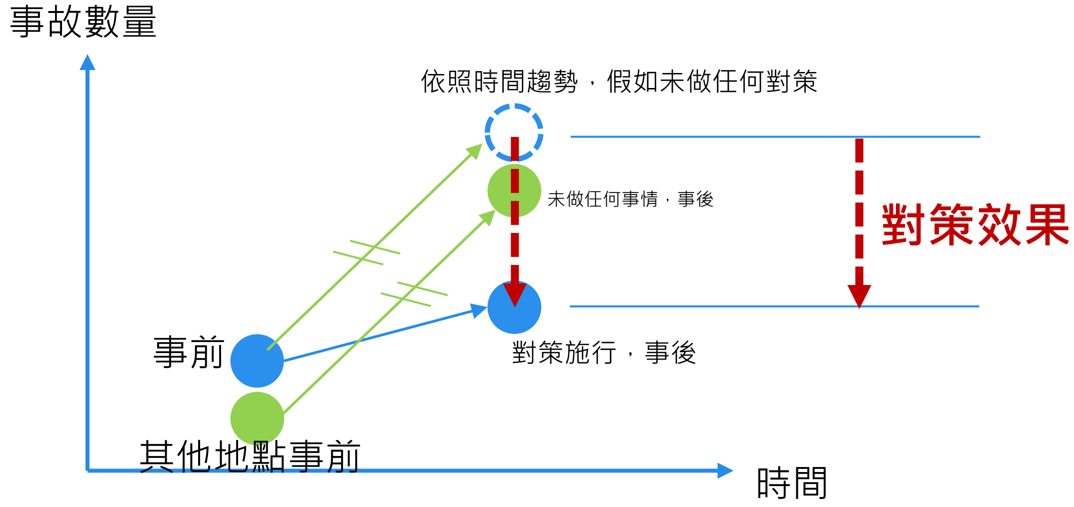
所以…「時間趨勢效應」

Time Trend Effects

- 和安慰劑效應有一樣的數學結構…

號誌化路口
安裝行人庇護島





線性、卜瓦松、負二項

$Y \sim$ Poisson　　$Y \geq 0$

$Z \sim$ Negative Binomial　　$Z \geq 0$

$Y_i = \Sigma \beta x$

- $y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2}$ ...
- 線性回歸的y值 可能為正的、可能為負的

$y = \beta [X]$

- 如果要求事故數量Y_i 永遠為正數
  - $Y_i = e^{\Sigma_j \beta_j x_{ij}} > 0$
  - $Y_i > 0$

$\exp(\Sigma \beta X)$ $e^{\Sigma \beta X}$

  - 可設定Y_i 符合朴瓦松(Poisson)分布，有平均值、變異數(Var)λ
  - Y_i ~ Poisson(λ)

$e^{-2} \geq 0$ $e^{+1.2} \geq 0$ $e^{-0.9} \geq 0$

- $Y_{i=1}$ ~ Poisson(λ) → $P\ \{\ Y_{i=1}\ = k_1\} = \lambda^{k_1} e^{-\lambda} /\ k_1!$
- $Y_{i=2}$ ~ Poisson(λ) → $P\ \{\ Y_{i=2}\ = k_2\} = \lambda^{k_2} e^{-\lambda} /\ k_2!$
- $Y_{i=3}$ ~ Poisson(λ) → $P\ \{\ Y_{i=3}\ = k_3\} = \lambda^{k_3} e^{-\lambda} /\ k_3!$
- …

也就是觀測到($y_{i=1}$, $y_{i=2}$, $y_{i=3}$) ... 的機率為：

$$P\ \{\ Y_{i=1}\ = y_1\}\ P\ \{\ Y_{i=2}\ = y_1\}\ P\ \{\ Y_{i=3}\ = y_1\}$$

- $Y_{i=1}$ ~ Poisson(λ) → $P\ \{\ Y_{i=1}\ = y_1\} = \lambda^{k_1} e^{-\lambda} /\ k_1!$
- $Y_{i=2}$ ~ Poisson(λ) → $P\ \{\ Y_{i=2}\ = y_2\} = \lambda^{k_2} e^{-\lambda} /\ k_2!$
- $Y_{i=3}$ ~ Poisson(λ) → $P\ \{\ Y_{i=3}\ = y_3\} = \lambda^{k_3} e^{-\lambda} /\ k_3!$

(續)

- 今天的狀況為，參數θ 不知道
  - L = $P\{Y_{i=1} = y_1\}\,P\{Y_{i=2} = y_2\}\,P\{Y_{i=3} = y_3\}\,\ldots$
  - L($\boldsymbol{\theta}$) = $\prod_i P\{Y_i = y_i|\theta\}$
  - L ($\boldsymbol{\theta}$) = $\lambda^{y_1+y_2+\cdots} e^{-\lambda-\lambda-\lambda\ldots} / [\,y_i!\,(y_2!)\ldots(y_i!)]$, $\boldsymbol{\theta}$=[λ]
  - L ($\boldsymbol{\theta}$) = $\prod_i \lambda^{k_i} \quad \prod_i e^{\lambda} / \prod_i \quad (k_i!)$
- Ln L($\boldsymbol{\theta}$) = $\Sigma_{\mathrm{i}}\, y_i + \ln(\lambda) - n\lambda \;- \Sigma_i ln(k_i!)$
- $\max_{\boldsymbol{\theta}}$ L($\boldsymbol{\theta}$) ← → $\max_{\boldsymbol{\theta}}$ Ln (L($\boldsymbol{\theta}$))

using the `glm()` function in R.

Poisson

$$\frac{Var(y_i)}{mean(y_i)} \approx 1$$

1.05 1.10 1.3?

- Poisson: Mean(yi) ~ Variance (yi)
- If Var(yi) > Mean(yi), use negative binomial
- If Var(yi) < Mean, use what distribution?

HW

- 嚴謹地判別，用什麼檢定(test)?

HW

- 若yi大多為0，用零膨脹朴瓦松(zero-inflated Poisson)

縱橫資料

Panel Data

縱與橫

Longitudinal and Cross-section Data

- 固定某個時間，各個地點之資料稱為橫斷面(cross section)資料
- 某個個體(路口/人)隨時間推移之資料，稱為「時間序列(time series)」，又稱縱段面(longitudinal data)

- 優點: 時間尺度可以自行整合。例如有某路口每個月的資料，連續3年
  - 114年1月、2月…12月；115年1月、2月…12月；116年1月…12月
  - 則這個路口可以有36個樣本
  - 或是整合為每年，則有3個樣本
  - 或是整合為每季，則有12個樣本… 取決於哪一個切分可以得到(統計、計量方面)更有效(efficient)的推論

“寬”資料

“Wide” Data

- 縱橫資料，一般紀錄為「寬資料」(wide data)
- 例如，五個路口的事故資料，在報表(spreadsheet)裡面往下為路口(橫斷面)，往右為依據每年推移(縱斷面)，如下：

事故數量

		110年											
		1月	2月	3月	4月	5月	6月	7月	8月	9月	10月	11月	12月
中興新村	CS0	0	0	0	0	0	0	0	0	0	1	0	1
潮州燒冷冰	CS1	6	1	2	1	1	2	2	1	3	1	1	3
歸仁	CS2	3	1	2	4	3	4	1	1	5	2	2	1
芒果	CS3	0	0	0	1	0	0	0	1	0	1	0	0
三多商圈	CS4	0	0	0	0	0	0	0	1	0	0	0	0
中山大學	CS5	0	0	0	0	0	0	0	0	0	1	0	1

		111年											
		1月	2月	3月	4月	5月	6月	7月	8月	9月	10月	11月	12月
中興新村	CS0	0	0	0	1	0	0	0	0	0	0	0	0
潮州燒冷冰	CS1	2	1	3	2	2	2	2	1	1	2	1	2
歸仁	CS2	6	2	4	4	1	4	2	2	2	X	X	X
芒果	CS3	0	0	0	0	0	0	1	0	0	1	0	0
三多商圈	CS4	0	0	0	0	0	0	1	0	0	0	0	0
中山大學	CS5	0	0	0	0	0	0	0	0	1	0	0	0

		112年											
		1月	2月	3月	4月	5月	6月	7月	8月	9月	10月	11月	12月
中興新村	CS0	0	0	0	0	0	0	0	0	0	0	0	0
潮州燒冷冰	CS1	4	4	X	X	X	X	X	3	3	1	0	2
歸仁	CS2	X	X	1	1	2	1	2	2	0	2	3	2
芒果	CS3	1	1	0	0	0	1	0	0	1	1	1	1
三多商圈	CS4	0	0	0	0	0	0	0	0	0	0	0	0
中山大學	CS5	0	0	0	0	0	0	0	0	0	0	0	0

		113年											
		1月	2月	3月	4月	5月	6月	7月	8月	9月	10月	11月	12月
中興新村	CS0	2	0	3	2	0	0	1	0	1	0	2	X
潮州燒冷冰	CS1	1	0	3	1	0	2	1	1	1	1	1	1
歸仁	CS2	2	2	3	1	1	1	0	2	1	1	2	2
芒果	CS3	1	0	0	1	0	0	0	0	X	X	X	X
三多商圈	CS4	X	X	X	X	0	0	0	0	0	0	0	0
中山大學	CS5	0	0	0	0	0	0	0	0	0	1	0	0

		114					
		1月	2月	3月	4月	5月	6月
中興新村	CS0	X	X	X	0	0	0
潮州燒冷冰	CS1	3	1	1	2	0	0
歸仁	CS2	4	1	0	3	0	0
芒果	CS3	X	0	0	1	0	0
三多商圈	CS4	0	0	0	0	0	1
中山大學	CS5	1	0	0	0	0	0

- 計量分析時，稍待會轉換為 “長資料”…

對策效果

- 以「雙元變數(binary variable)」表達某個時間某個地點是否已改善(treated)。
- 如果時間點t，路口i為改善後，則 $D_{i,t}$ = 1 ，否則為0
- 例：現代圓環，數個地點，假如以月為單位紀錄：

芒果圓環113年9月施工，113年12月完工，兩個月適應期，114年2月開始為事後

$D_{芒，113.06}=0$　$D_{芒，113.07}=0$　$D_{芒，113.08}=0$

$D_{芒，114.02}=1$　$D_{芒，114.03}=1$　$D_{芒，114.04}=1$

雙元變數(binary variable)，又稱虛擬變數(dummy variable, 啞變數)

縱橫資料建模

Panel Data Construction

- $Y_{i,t} = \beta_0 + \underbrace{\alpha_i \cdot F_i}_{\substack{Fixed\ Effect \\ (absorbing\ traffic\ volumes)}} + \gamma \cdot D_{i,t} + \sum \beta_t \cdot \delta_t + \epsilon_{i,t}$

- **所求**: 某一個地點，**同一個時間點**，有做改善，**對照「假如沒做」**
  - 「假如沒做」，與現實不符(counterfactual)，如同平行宇宙(parallel universe)，但可由模型抓出。
  - 所求 = $(y_{i,t} | D_{i,t} = 1) - (y_{i,t} | D_{i,t} = 0)$

$$\left\{\beta_0 + \underbrace{\alpha_i \cdot F_i}_{\substack{Fixed\ Effect \\ (absorbing\ traffic\ volumes)}} + \gamma \cdot 1 + \sum \beta_t \cdot \delta_t + \epsilon_{i,t}\right\} - \left\{\beta_0 + \underbrace{\alpha_i \cdot F_i}_{\substack{Fixed\ Effect \\ (absorbing\ traffic\ volumes)}} + \gamma \cdot 0 + \sum \beta_t \cdot \delta_t + \epsilon_{i,t}\right\}$$

$$= \gamma \cdot (1 - 0)$$

$$= \gamma$$

#

時間因子

Time Factors

- $Y_{i,t} = \beta_0 + \underbrace{\alpha_i \cdot F_i}_{\substack{Fixed\ Effect \\ (absorbing\ traffic\ volumes)}} + \gamma \cdot 1 + \sum \beta_t \cdot \delta_t + \epsilon_{i,t}$
- δ_t為雙元變數(虛擬變數)，數據等於某年，則δ_t=1，否則為0。其所造成的效果為所帶係數β_t。
- 例如，110年的資料，$\delta_{111} = 0$；$\delta_{112} = 0$；$\delta_{113} = 0$；$\delta_{114} = 0$
  - 111年的資料，$\delta_{111} = 1$；$\delta_{112} = 0$；$\delta_{113} = 0$；$\delta_{114} = 0$
  - 112年的資料，$\delta_{111} = 0$；$\delta_{112} = 1$；$\delta_{113} = 0$；$\delta_{114} = 0$
  - 113年的資料，$\delta_{111} = 0$；$\delta_{112} = 0$；$\delta_{113} = 1$；$\delta_{114} = 0$
  - 114年的資料，$\delta_{111} = 0$；$\delta_{112} = 0$；$\delta_{113} = 0$；$\delta_{114} = 1$
- 如同先前世代變數，各個地點經歷的時間，可能有共同的高低起伏
  - 例如大多數國家新冠疫情時，(2020年，109年) 事故下降
  - 因此$Y_{i,t=109}$可能較其他年份低，若此假設為真，將由模型係數$\beta_{109} < 0$ 反映出來

固定效果

- $$Y_{i,t} = \beta_0 + \underbrace{\alpha_i \cdot F_i}_{\substack{Fixed\ Effect \\ (absorbing\ traffic\ volumes)}} + \gamma \cdot 1 + \sum \beta_t \cdot \delta_t + \epsilon_{i,t}$$
- F_i為雙元變數， F_i=1表示地點i，反之為F_i=0。特定地點(site-specific)。
- 每個地點都有自己的特色，例如黃昏眩光、車流量大...無法一一納入，因此用一個固定效果(fixed effect)吸收(absorbed)，推算較高之$Y_{i,t}$.
- 假如某路口 i 就是因為特定時候眩光、又車流量大 →特定地點(site-specific)較多事故，效果為 $\alpha_i > 0$
  - 本路口 $\alpha_i \cdot 1 = \alpha_i$
  - 其他路口 $\alpha_i \cdot 0 = 0$

“寬”轉換成 “長”資料

PART IV: 左轉車道案例

計畫背景

https://myway.cpami.gov.tw/qualityup_highlight_page/20

苗栗縣亮點計畫-漫步苗北雙城-頭份竹南城市慢行綠網建置計畫

https://youtu.be/SZu2FSBunOQ?si=96cN5z8lvviJlheu

左轉車道是否有效

Issues Need to be Addressed

所以，有效(valid)的量化分析，須解決

- 1. **回歸趨中**(Regression-to-the-mean)
    - Selection bias
    - Not sampling average

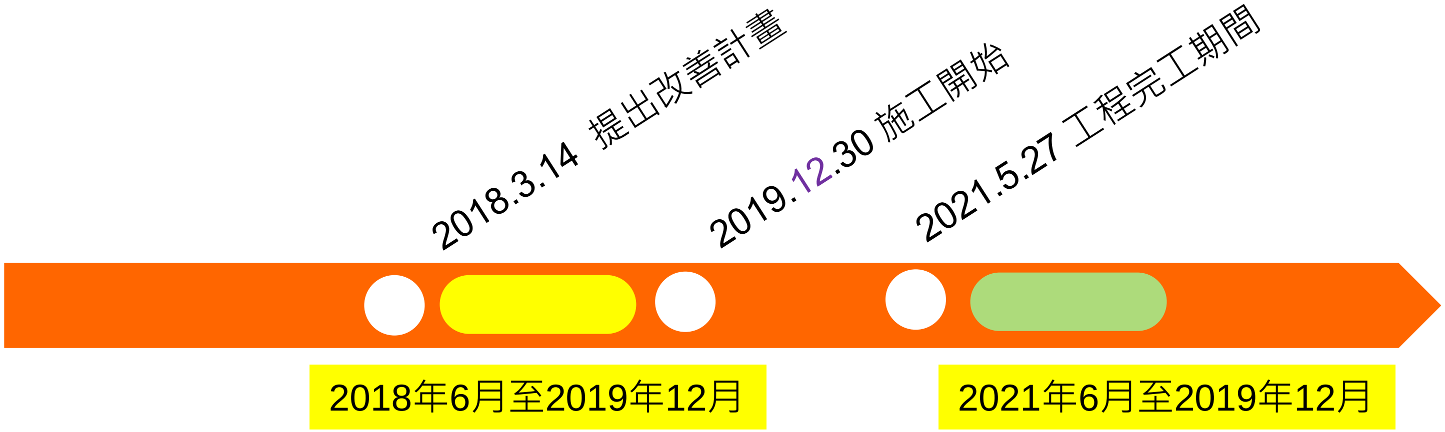
- 2. **時間趨勢效應**(Time trend effects) (El-Basyouny & Sayed, 2011; Guo & Sayed, 2020)

免疫時間窗—壓制選擇偏誤

Immune Window - Managing “Selection Bias”

免疫時間窗

- 免疫時間窗(Wei et al., preparing)：計畫核定後、動工前的一小段時間，可免於趨中回歸(RTM)的影響



※2018年3月14日: 漫步苗北雙城-頭份竹南城市慢行綠網建置計畫於2018年3月14日

- 免疫時間窗 EXCEL 檔練習

Odds Ratio-處理大趨勢

- 製藥業：的對策效果：看差值
    - $E = [\,(\bar{y}_{\text{after}} |\, D = 1) \;-\; (\bar{y}_{\text{before}} |\, D = 1)] - [(\bar{y}_{\text{after}} |\, D = 0) \;-\; (\bar{y}_{\text{before}} |\, D = 0)]$
- 道路交通安全領域： 看百分比
    - 先求「比值(ratio)」
    - 再求「百分比」

	差值	比值
樸素(不可信)	(B - A)	B/A
成熟	[(B-A) – (C- A’)]	(B/A) / (C/A’)

$$\text{OddsRatio } OR = \frac{[\,(\bar{y}_{\text{after}} | D{=}1)/(\bar{y}_{\text{before}} |\, D{=}1)]}{[(\bar{y}_{\text{after}} |\, D{=}0)\;/(\bar{y}_{\text{before}} |\, D{=}0)]}$$

- 如果事故為Poisson分布
  - y_i ~ Poisson
  - $y_i = \exp\left(\beta_0 + \beta_1 T_t + \beta_2 D_i + \delta\, T_t \times D_i + \ \Sigma_j \beta_j x_{ij}\right)$

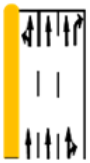
OddsRatio $OR \ = \dfrac{[\,(\bar{y}_{\text{after}}|D{=}1)/(\bar{y}_{\text{before}}|\,D{=}1)]}{[(\bar{y}_{\text{after}}|\,D{=}0)\ /(\bar{y}_{\text{before}}|\,D{=}0)]}$

$OR =$

有工程概念的建模

Model Building

分解為主線連續性、主線順暢性

	Left-turn trap	Shared LT+TH	Additional LT Lane		
			Without lateral shift	Partial lateral shift	Full lateral shift
Example					
Mainstream Continuity *C*	0	1	1	1	1
Mainstream Smoothness *S*	0	0	1	1	1
Lateral Shift *L*	N/A	N/A	0	Between 0 and 1	1
Effective Lateral Shift *L'*	0	0	0	Between 0 and 1	1

Source: Wei et al. (Preparing)

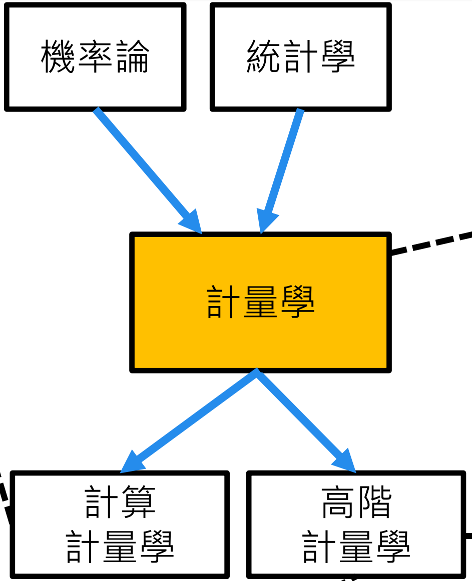
車流量為曝光量

事故與{流量開根號}正比

- $Y \propto \sqrt{V}$

- $Y_i = \sqrt{V_i}e^{\Sigma_j \beta_j x_{ij}} = e^{\ln(\sqrt{V_i})} e^{\Sigma_j \beta_j x_{ij}} = e^{\gamma' \ln(\sqrt{V_i}) + \Sigma_j \beta_j x_{ij}}$, γ'=1
- Or $Y_i = e^{\gamma \ln(V_i) + \Sigma_j \beta_j x_{ij}} = e^{\ln(V_i)^{\gamma} + \Sigma_j \beta_j x_{ij}}$, γ~(1/2) or other value

課程地圖



課程資訊	
課程名稱	計量經濟學計算方法 Computational Methods for Econometrics
開課學期	108-2
授課對象	社會科學院 經濟學研究所
授課教師	謝志昇
課號	ECON7218
課程識別碼	323EM3770
班次	
學分	3.0
全/半年	半年
必/選修	選修
上課時間	星期一2,3,4(9:10~12:10)
上課地點	社科401
備註	本課程以英語授課。 限碩士班以上 總人數上限：20人
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖
課程大綱	
為確保您我的權利,請尊重智慧財產權及不得非法影印	
課程概述	In modern economic research, computers enhance our capacity of analyzing complex problems with data support. Computation is particularly important in fields involving dynamic modeling, structural equations, and massive data, such as macro, labor, and industrial organization. However, computational methods have not been part of the core curriculum of postgraduate-level economics education, whereas programming skills are critical for a postgraduate's success in academia and industry. The objective of this course is to introduce graduate students to commonly applied computational approaches for solving econometric models, with an emphasis on numerical optimization, Bayesian MCMC, simulation-based methods, and dynamic programming. We expect that at the end of the course a student would proficiently use at least one programming language (Stata, Matlab, R, etc). Moreover, we aim to equip the students with the computational ability to tackle problems of their own research areas.

1. Introduction
2. Constrained estimation (Ch.8 of Hansen's (2020) textbook)
   - Minimum distance (MD) estimator, asymptotic distribution of the MD, asymptotically efficient MD estimator, misspecification and local misspecification.
3. Hypothesis testing and model selection (Ch.9, Ch.28)
   - Hypotheses, type I error, type II error, significance, p-value, Wald test, score test, criterion-based test, testing consistency, asymptotic local power, information criterion, AIC and BIC.

1

4. Bootstrap (Ch.10)
   - Bootstrap algorithm, bootstrap moments, bootstrap distribution, bootstrap asymptotics, bootstrap consistency, bias correction, bootstrap tests, parametric bootstrap, bootstrap regression, and wild bootstrap.
5. Endogeneity and IVs (Ch.12)
   - Endogenous regressors, identification, two-stage LS (2SLS) and properties, generated regressors, control function, endogeneity tests, weak IVs.
6. GMM (Ch.13)
   - Moment equation models, method of moments, overidentified moment equations, GMM estimator, efficient GMM, overidentification test, conditional moment equation models.

– Midterm exam. 4/22

7. Time series data (Ch.14)
   - Stationarity, convergent series, ergodicity, martingale difference, mixing, linear projection, white noise, Wold decomposition, lag operator, unit root.
8. Univariate time-series models (Ch.14)
   - Moving average (MA) processes, autoregressive (AR) processes, ARMA processes, ARIMA processes, time trend, time-series regressions, estimation, asymptotics.
9. Multivariate time-series models (Ch.15)
   - Linear projection, impulse response, VAR models, estimation, asymptotics, orthogonalized shocks, forecast error decomposition, recursive VAR, structural VAR.
10. Non-stationary time series (Ch.16)
   - Partial sum, functional convergence, functional CLT, integration order, unit-root tests, spurious regression, non-stationary VAR, cointegration, cointegrated VAR.

課程資訊	
課程名稱	計量經濟學二 Econometrics (II)
開課學期	109-2
授課對象	管理學院 財務金融學研究所
授課教師	陳宜廷
課號	Fin8037
課程識別碼	723ED6100

課程資訊	
課程名稱	計量研究與應用 Applied Econometrics
開課學期	111-2
授課對象	工學院 營建工程與管理組
授課教師	荷世平
課號	CIE7135
課程識別碼	521 M7230
班次	
學分	2.0
全/半年	半年
必/選修	選修
上課時間	星期二B,C(19:20~21:05)
上課地點	土研405
備註	限碩士在職專班生 總人數上限：30人
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖
課程大綱	目前無該大綱資料！

1.The nature of Econometrics and economic data
2.The simple regression model
3.Multiple regression analysis: Estimation
4.Multiple regression analysis: Inference
5.Multiple regression analysis: OLS asymptotics and further issues
6.Multiple regression analysis with qualitative information
7.Heteroskedasticity
8.Specification error and data problems
9.Basic regression analysis with time series data
10.Further issues in using OLS with time series data
11.Serial correlation and heteroskedasticity in time series regressions
12.Pooling cross sections across time (Panel data method)
13.Advanced panel data methods
14.Instrumental variables estimation and two stage least squares
15.Simultaneous equations models
16.Limited dependent variables models and sample selection corrections
17.Advanced time series topics

課程資訊	
課程名稱	計量經濟學的因果推論與預測 Causal Inference and Prediction in Econometrics
開課學期	109-1
授課對象	社會科學院 經濟學系
授課教師	郭濡豪
課號	ECON5179
課程識別碼	323EU4300
班次	
學分	2.0
全/半年	半年
必/選修	選修
上課時間	星期二3,4(10:20~12:10)
上課地點	社科402
備註	本課程以英語授課。 限學士班三年級以上 或 限碩士班以上 或 限博士班 總人數上限：50人
Ceiba 課程網	

週次	日期	單元主題
第1週	9/15	Meanings of identification in econometrics
第2週	9/22	Meanings of identification in econometrics
第3週	9/29	Moment-based identification and estimation
第4週	10/06	Moment-based identification and estimation
第5週	10/13	Extremum-based identification and estimation
第6週	10/20	Extremum-based identification and estimation
第7週	10/27	Endogeneity problems in econometrics
第8週	11/03	Endogeneity problems in econometrics
第9週	11/10	Midterm examination
第10週	11/17	Model selection and prediction
第11週	11/24	Model selection and prediction
第12週	12/01	Causality in econometrics
第13週	12/08	Causality in econometrics
第14週	12/15	Frequentist, Bayesian, and Fisherian inferences
第15週	12/22	Frequentist, Bayesian, and Fisherian inferences
第16週	12/29	Frequentist, Bayesian, and Fisherian inferences
第17週	1/05	Frequentist, Bayesian, and Fisherian inferences

課程資訊	
課程名稱	類別資料分析 Categorical Data Analysis
開課學期	108-2
授課對象	理學院 心理學系
授課教師	姚開屏
課號	Psy5011
課程識別碼	227 U0300
班次	
學分	3.0
全/半年	半年
必/選修	選修
上課時間	星期二2,3,4(9:10~12:10)
上課地點	南館地下A
備註	總人數上限：20人
Ceiba 課程網頁	http://ceiba.ntu.edu.tw/1082Psy5011_CDA
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖

週次	日期	單元主題
第1週	03/03	Introduction, Chap. 1
第2週	03/10	CH2 Two-Way & Three-Way Contingency Tables
第3週	03/17	CH3 Generalized Linear Models
第4週	03/24	CH3 Generalized Linear Models
第5週	03/31	CH4 Logistic Regression
第6週	04/07	CH4 Logistic Regression
第7週	04/14	CH5 Building & Applying Logistic Regression Models
第8週	04/21	CH5 Building & Applying Logistic Regression Models
第9週	04/28	CH6 Multicategory Logit Models
第10週	05/05	CH6 Multicategory Logit Models
第11週	05/12	CH7 Loglinear Models
第12週	05/19	CH7 Loglinear Models
第13週	05/26	CH8 Models for Matched Pairs

HW(有些為加分)

統計

- Chow's Test
- Tuckey Test
- Walt Test, Likelihood ratio test…

計量

- Granger Test 測因果關係
- 2SLS
- Jack knife
- Instrument Variable (IV)
- Seemly Unrelated Regression, SUR

Wald, A. (1943). Tests of statistical hypotheses concerning several parameters when the nu mber of observations is large. *Transactions of the American Mathematical society*, *54*(3), 426 -482.

Granger, C. W. (1969). Investigating causal relations by econometric models and cross-spectral methods. Econometrica: journal of the Econometric Society, 424-438.

實務重要性

「實務重要性」>>「統計顯著性」

Practical Importance >> Statistical Significance

- CROW(2009), p.156

If a result is statistically significant, this does not automatically mean that the result is also of practical importance. Significance should not be confused with importance. Whether a particular result is significant largely depends on the size of the sample. The larger the sample, the more likely it is that the result will be significant. In large-scale samples, differences that have little meaning can be significant nonetheless. Should, for example, value be attached to a 0.2% reduction in the percentage of people who wear their seat belts if this is measured in a sample of 100,000 motorists? Significant simply means that a result is not founded on coincidence. The question is whether practical consequences should always be attached to significant differences (for example set up a campaign focusing on the importance of wearing a seat belt) [4.19].

什麼是統計顯著

- 統計上顯著的意思，通常測試一個欲推測的值，是否「與0相異」
  - 所以，統計上顯著多活4天，如果p-value為0.03，則只是表97%的機會這個取樣造成的結果不是偶然不同於0
  - 但一直吃某藥 但多活4天，是否有實務重要性?

主成分分析(PCA)

主成分分析(PCA)

Principal Component Analysis

課程大綱 | 課程大綱 | 課程大綱 | 課程大綱 | Principal Comp | 課程大綱

google.com/search?q=Principal+Component+Analysis,+nol.ntu.edu.tw&sca_esv=3ffb8f4f6ca8e7f9&sxsrf=ACQVn0_ssaUVcKdK...

Google

Principal Component Analysis, nol.ntu.edu.tw

All | Images | Videos | Shopping | News | More | Tools

Past year | All results | Clear

臺大課程網
https://nol.ntu.edu.tw › print_table · Translate this page

資料科學方法論

Oct 9, 2023 — The aim of this course is to introduce a variety of Statistical and Machine Learning data **analysis** methods. Three core techniques for data science: Data ...

臺大課程網
https://nol.ntu.edu.tw › print_table · Translate this page

機器學習與環境資料分析

5 hours ago — **No**. 項目. 百分比. 說明. 1. Final project ... (14) Unsupervised Learning: **Cluster analysis**, Self-organizing maps, **Principal component analysis** {*Presentation: 14.5}.

國立臺灣大學資訊工程學系
https://www.csie.ntu.edu.tw › ~htlin › course

Hsuan-Tien Lin > Courses > Machine Learning, Fall 2023

Jan 6, 2024 — This course introduces the basics of learning theories, the design and **analysis** of learning algorithms, and some applications of machine learning. People.

課程資訊	
課程名稱	資料科學方法論 Computational Methods for Data Science
開課學期	111-2
授課對象	電機資訊學院 資料科學碩士學位學程
授課教師	張明中
課號	Data5010
課程識別碼	946 U0100
班次	
學分	3.0
全/半年	半年
必/選修	選修
上課時間	星期三6,7,8(13:20~16:20)
上課地點	新504
備註	限碩士班以上 總人數上限：30人
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖
課程大綱	
為確保您我的權利,請尊重智慧財產權及不得非法影印	
課程概述	第一週 Introduction to Data Science and Matrix Algebra 第二週 Data Collection: Survey sampling 第三週 Data Collection: Factorial design and Space-filling design 第四週 Data Analysis: Supervised Learning I – Linear model and Generalized linear model 第五週 Data Analysis: Supervised Learning II -- Nonparametric regression 第六週 Data Analysis: Supervised Learning III -- Gaussian process regression 第七週 Data Analysis: Supervised Learning IV -- Discriminate analysis 第八週 Data Analysis: Supervised Learning V -- Support vector machine 第九週 期中考週 第十週 Data Analysis: Supervised Learning VI -- Bagging, Random forests, Boosting 第十一週 Data Analysis: Supervised Learning VII -- Deep neural networks 第十二週 Data Analysis: Unsupervised Learning I – Principal component analysis, Factor analysis, Canonical correlation analysis 第十三週 Data Analysis: Unsupervised Learning II -- Clustering methods 第十四週 Big Data Issue I (p>>n): Feature screening 第十五週 Big Data Issue II (n>>p): Subdata selection 第十六週 期末考週 第十七週 彈性教學 第十八週 彈性教學

Machine Learning, Fall 2023

Course Description

Machine learning allows computational systems to adaptively improve their performance with experience accumulated from the data observed. This course introduces the basics of learning theories, the design and analysis of learning algorithms, and some applications of machine learning.

People

- instructor: Hsuan-Tien LIN (htlin AT csie . ntu . edu . tw) [office hour: after classes, or by appointment]
- TAs and TA hour: html_ta AT csie . ntu . edu . tw
  - Chia-Wei CHANG (Undergraduate in CSIE Department)
  - Yu-Cheng CHENG (M.S. Student in CSIE Department)
  - Shuo-Chen HO (Undergraduate in CSIE Department)
  - Cai-Yi HU (M.S. Student in CSIE Department)
  - Yu-Shiang HUANG (Ph.D. Student in Data Science Program)
  - Ren-Wei (Willy) LIANG (Undergraduate in CSIE Department)
  - Jeng-Yue (Buffett) LIU (Undergraduate in Geography Department)
  - Poy LU (Ph.D. student in Graduate Institute of Networking and Multimedia)
  - Odo To (Undergraduate in CSIE Department)
  - Cheng-Chi (Casper) WANG (Undergraduate in CSIE Department)
  - Hsuan-Fu WANG (M.S. Student in Graduate Institute of Networking and Multimedia)

Course Information

- Time: Wednesdays 9:10 to 12:10
- Room: CSIE R104 (with broadcast to R102)
- NTU COOL: https://cool.ntu.edu.tw/courses/32477
- Slido: #HTML2023FALL
- Textbook: *Learning from Data*, by Yaser Abu-Mostafa, Malik Magdon-Ismail and Hsuan-Tien Lin
- Language: **Mandarin** teaching
- Grading: 70% homework, 30% project (tentative)

12/06 (W14)	**topic 7: how can machines learn by distilling hidden features?** neural network; (preliminary) deep learning	homework 5 due	**required** watching (before class): • course slides; LFD e-7.1, e-7.2, e-7.3, e-7.4 (selected parts) • Neural Network :: Motivation • Neural Network :: Neural Network Hypothesis • Neural Network :: Neural Network Learning • Neural Network :: Optimization and Regularization **required** watching (before class): • course slides; LFD e-7.6 • Deep Learning :: Deep Neural Network • Deep Learning :: Autoencoder • Deep Learning ::Denoising Autoencoder • Deep Learning :: Principal Component Analysis
			suggested watching:

(續)

評論：

- 「大數據」才需要壓縮
- 普通大小的資料→沒必要用PCA

時間序列

ARIMA

- AR
- MA
- ARMA
- ARIMA

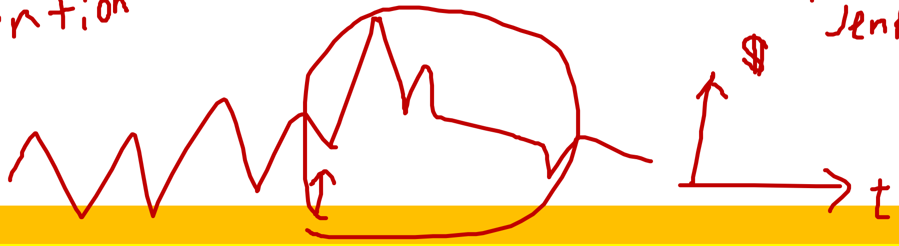
T=1	Y(t=1)
T=2	Y(t=2)
T=3	…
…	…
…	…

廣告

"intervention"

G.C. Tiao

Box Jenkins



課程資訊	
課程名稱	計量經濟專題討論 Special Topics in Econometrics
開課學期	100-1
授課對象	生物資源暨農學院 農業經濟學研究所
授課教師	黃芳玫
課號	AGEC7028
課程識別碼	627 M1790
班次	
學分	3
全/半年	半年
必/選修	選修
上課時間	星期一2,3,4(9:10~12:10)
上課地點	農經二
備註	與林金龍合開 總人數上限：30人
Ceiba 課程網頁	http://ceiba.ntu.edu.tw/1001ste
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖

課程進度

週次	日期	單元主題
第2週	9/19	Conditional Expectations
第3週	9/26	Basic Asymptotic Theory
第4週	10/03	Introduction to time series analysis and R
第5週	10/10	雙十節國慶日放假
第6週	10/17	ARIMA modeling I/ identification
第7週	10/24	ARIMA modeling II/ estimation and diagnostic checking
第8週	10/31	ARIMA modeling III/ Forecasting
第9週	11/07	Linear model I
第10週	11/14	Midterm Exam
第11週	11/21	Instrumental Variables I
第12週	11/28	Instrumental Variables II
第13週	12/05	Panel Data I
第14週	12/12	Panel Data II
第15週	12/19	Panel data III
第16週	12/26	Treatment Effect II
第17週	1/02	Treatment Effect III
第18週	01/09	Final Exam

縱橫資料

Panel Data

Panel Data

Panel Data

- 「縱斷面(longitudinal) + 橫斷面(cross-sectional)」
- 時間序列 + 複數個體

	i=1	i=2	i=3
T=1	Y(1,1)	Y(2,1)	Y(3,1)
T=2	Y(2,1)	…	…
T=3	…	…	…
T	…	Y(2,t)	…
…	…	…	…

寬→長資料

Wide → Long

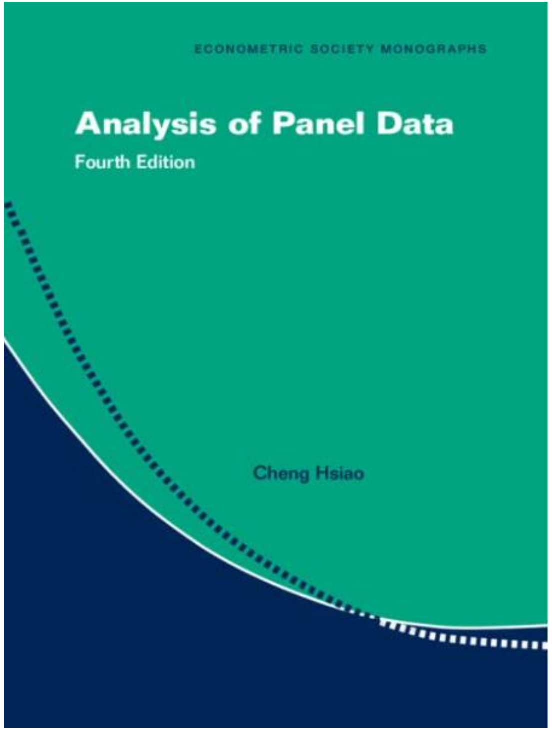
案例(上機)

- 現代圓環的對策效果(Treatment effect) 是什麼？

- (0) 下載R-Markdown
- (1) 開啟表格(這是Long.實務上要先做成Wide肉眼比較好確認)
- (2)輸入Panel Data 的回歸R語法：

- (3)看效果。但尚未考慮對照組，不夠準

Panel Data(Cont'd)



Hsiao

資源

- 教科書:Hsiao
- 台大課程

課程資訊	
課程名稱	計量經濟學 Econometrics
開課學期	104-1
授課對象	管理學院 國際企業學研究所
授課教師	李 揚
課號	IB8063
課程識別碼	724 D6500
班次	
學分	3
全/半年	半年
必/選修	必修
上課時間	星期四7,8,9(14:20~17:20)星期五2,3,4(9:10~12:10)
上課地點	
備註	博士班國企組、策略組、行銷組必修。週四、週五單週上課。上課教室：管一401。週四單週上課。週五單週上課。 限博士班 總人數上限：20人
Ceiba 課程網頁	http://ceiba.ntu.edu.tw/1041IB8063_
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖
課程大綱	
為確保您我的權利,請尊重智慧財產權及不得非法影印	
課程概述	TOPICS 0. Basic knowledge of statistics 1. Matrix algebra 2. Ordinary least squares (OLS) estimation of the simple regression model: Estimation, tests, and prediction. 3. OLS estimation of the multiple regression model 4. Multiple Regression in Matrix Form 5. Extensions of the multiple regression model: Dummy independent variables 6. Generalized least square (GLS): Heteroscedasticity and serial correlation 7. Discrete regression model 8. Limited dependent variable 9. Instrumental variables 10. Panel data 11. SUR and 3SLS

「控制組」的重要性

CROW課本探討

- 計量分析
  - 理想上要有
    - 時間區別：事前(before/pretest)、事後(after/posttest)
    - 實驗組別：對策組(treatment group)、控制組(control group)
  - 可明確抓到大趨勢
- 有時候，沒有事前資料：
    - 「只有橫斷面(cross-sectional)」，對策組(treatment group)、控制組(control group)
    - 因為對策前恰好還沒有資料
  - 仍看得出效果
- 有時候，沒有「控制組」
    - 只有縱斷面──事前(before/pretest)、事後(after/posttest)
    - → 不恰當! 不知是否為大趨勢造成

缺「事前」

Lack of ***Pretest***(before data)

4.5.4 Alternative designs and consequences for the conclusions

A number of alternative studies will now be dealt with that can also be used to answer the above-mentioned research question. We will look at how they differ from the ideal design and what the consequences are for the conclusions that can be drawn from the research results.

Experiment without pretest

The first alternative study design is identical to that dealt with in section 4.5.2, with the only difference being that no pretest is carried out (table 4.7). The only independent variable in this study design is 'condition'. The mean number of fatalities and serious injuries in the posttest was 9.2 for the experimental group and 20.1 for the control group.

Table 4.7. No pretest

		Posttest
Experimental group	s1	8
	s2	2
	...	...
	s30	12
Control group	s31	17
	s32	25
	...	...
	s60	26

Table 4.8. ANOVA table for the data in table 4.7

Source	SS	df	MS	F	P
1 condition	1760.4	1	1760.4	[1/2] = 76.4	0.000
2 error	1337.2	58	23.1		
3 total	3097.6	59			

As is clear from table 4.8, the test for 'condition' is significant at 1% level ($p < 0.01$), which means that the zero hypothesis for equal means in the experimental and the control groups can be rejected.

Can it now be concluded that the measure has led to a reduction in the number of casualties? The answer is yes, but only if the sixty intersections were randomly allocated to the experimental and control groups. If this was not the case, an alternative explanation for the impact that the measure was found to have had is that the experimental group and the control group differ in other aspects that also relate to the dependent variable. If, for example, the traffic volume over a 24-hour period at the intersections in the control group is coincidentally much higher than on the roundabouts in the experimental group, these differences form a potential alternative explanation for the effect found. In the analysis, such a characteristic can be statistically adjusted with the help of a co-variance analysis if this characteristic was also measured.

缺「控制組」

Lack of “Control Group”

Experiment without control group
Another alternative study design is shown in table 4.9, and features pretest and posttest but no control group.

The only independent variable now, therefore, is ‘time’. The average number of casualties during the pretest was 20.3 and in the posttest 9.2. The result of the variance analysis of this data is shown in table 4.10. The test for ‘time’ is significant at 1% level ($p < 0.01$), which means it can be concluded that the conversion of the thirty intersections into roundabouts is linked to the reduction in the number of traffic casualties.

The question now is whether it can be concluded that the measure brought about a reduction in the number of traffic casualties? The answer is no, because the reduction could just as easily have been caused by other measures such as:
1 a generally declining trend;
2 regression to the mean;
3 changes in registration;
4 external circumstances other than the measure.

A combination of these four alternative explanations is also possible. This is briefly explained below.
The first relates to the fact that the number of fatalities and serious traffic injuries in the Netherlands has been gradually declining since the 1970s. This could constitute a (partial) explanation for the differences between the pretest and posttest in the first alternative explanation above. The second explanation is relevant if the thirty intersections studied in the pretest phase show an above-average number of casualties. In this case, the number of casualties in the second measurement could have shifted towards the mean on the basis of coincidence alone, which in itself results in an apparent reduction in the number of casualties. This phenomenon is called statistical regression to the mean. In the third explanation, the number of casualties could only have apparently decreased if the police are less accurate in registering accidents in the posttest phase

Table 4.9. No control group

		Pretest	Posttest
Experimental group	s1	17	8
	s2	22	2
	...	...	...
	s30	24	12

Table 4.10. ANOVA table for the data in table 4.9

Source	SS	df	MS	F	P
1 time	1826.0	1	1826.00	[1/2] = 94.6	0.000
2 error(1)	559.5	29	19.3		
3 error(2)	687.8	29			
4 total	3073.3	59			

Record 26: Road safety manual

166

有人怪老師講錯

- “北方之橋，施欄楯以防失足而已。閩中多雨，皆於橋上覆以屋，以庇行人。邱二田言，有人夜中遇雨，趨橋屋坐。有一吏攜案牘，與軍役押數人避屋下。枷鎖瑯然，知為官府錄囚，懼不敢近，但畏縮於一隅中。一囚號哭不止，吏叱曰：「此時知懼，何如當日勿作耶？」囚泣曰：「吾為吾師所誤也。吾師日講學，凡鬼神報應之說，皆斥為佛氏之妄語。吾信其言，竊以為機械能深，彌縫能巧，則種種惟所欲為，可以終身不敗露。百年之後，氣返太虛，冥冥漠漠，並毀譽不聞，何憚而不恣吾意乎？不虞地獄非誣，冥王果有，始知為其所責，故悔而自悲也。」一囚曰：「爾之墮落由信儒，我則以信佛誤也。佛家之說，謂雖造惡業，功德即可以消滅；雖墮地獄，經懺即可以超度。吾以為生前焚香佈施，歿後延僧持誦，皆非吾力所不能，既有佛法護持，則無所不為，亦非地府所能治。不虞所謂罪福，乃論作事之善惡，非論捨財之多少。金錢虛耗，舂煮難逃，向非恃佛之故，又安敢縱恣至此耶？」語訖長號。諸囚亦皆痛哭。乃知其非人也。夫六經具在，不謂無鬼神；三藏所談，非以斂財賂。自儒者沽名，佛者漁利，其流弊遂至此極。佛本異教，緇徒藉是以謀生，是未足為責。儒者亦何必乃爾乎？” ——紀曉嵐閱《閱微草堂筆記 卷十四 槐西雜志四》

※新加坡機車庇護

※高雄機車庇護

參考文獻

教科書

- CROW (2009), “Road Safety Manual”

計量經濟學

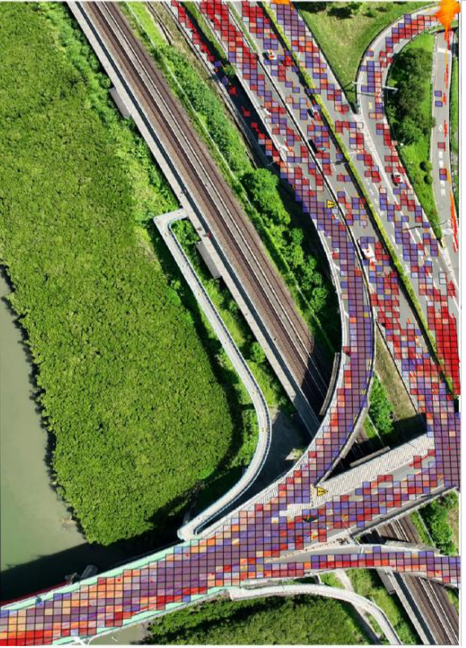
- Greene (2018) Econometric Analysis, 8th edition (India Subcontinent version)
- Angrist, J. D., & Pischke, J. S. (2009). Mostly harmless econometrics: An empiricist's companion. Princeton university press.
- Washington, S., Karlaftis, M. G., Mannering, F., & Anastasopoulos, P. (2020). *Statistical and econometric methods for transportation data analysis*. Chapman and Hall/CRC.

道路安全的混淆因子(confounding factors)

- Elvik, R. (2002). The importance of confounding in observational before-and-after studies of road safety measures. *Accident Analysis & Prevention*, *34*(5), 631-635.

左轉車道佈設

- Wei, S.-Y., Hsu, T.-P., Chen, Y.-H., Tsai, H.-Y.(Preparing) The Safety Effects of Left-turn Installation with Various Types – An Engineering-based Regression Analysis
- 魏三雅(2024)，佈設左轉專用車道之對於號誌化路口安全之影響，國立臺灣大學碩士論文



謝謝

Yenchen.ntu@gmail.com

來源:Google Map
@57.7276442,11.9719096

附錄

實用工具

Useful Tools

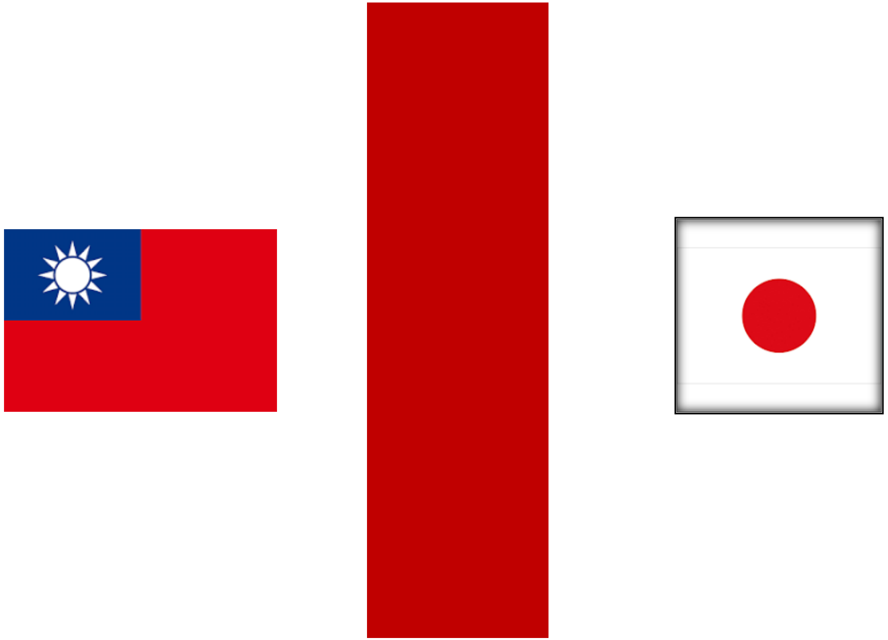
統計

- https://www.statskingdom.com/

DuckDuckGo

DuckDuckGo — Privacy, simp
duckduckgo.com
DuckDuckGo
擺脫追蹤，盡情搜尋
厭倦上網時被追蹤嗎？我們可以助你一臂之力。
切換至 DuckDuckGo，即可免費獲得萬無一失的隱私保護：
比無痕模式更私密
Private Search
終身免費
在 Windows下載 DuckDuckGo
將 DuckDuckGo 設定為預設瀏覽器
深受全球千萬用戶信賴！

- Google 以前可輕易搜尋到日文資訊
- Google內部疑似有境外勢力惡意阻擋




右折車線 at DuckDuckGo
duckduckgo.com/?t=h_&q=右折車線&ia=web
右折車線
全部
圖像
影片
新聞
地圖
設定
臺灣
安全搜尋： 適中
隨時
https://221616.com › norico › right-turn
「右折が怖い!」手順、待つ場所、優先順位、気を付けたいポイント...
右折車; そのため対向車線を直進してくるクルマに道を譲るのはもちろん、対向車線から左折するクルマがいれば、そのクルマにも道を譲りましょう。 もちろん曲がった先にある横断歩道を通る歩行者がいれば、その歩行者が優先です。
右折車線 的影片
2:30
右折の基本を徹底解説
右折
どこを見る？
4:57
二段階右折禁止の場合の右折の仕方！
6:29
【危険予知トレーニング】交差点編 #13 ～右折車線が２レーンある
102K views
YouTube | 1年
【右折が苦手な方必見】右折の基本を徹底解説!
311K views
YouTube | 3年
初心者講座８：原付の右折方法【リクエスト動画】
328K views
YouTube | 8年
更多影片
https://driving.ciao.jp › 安全に右折するための完全ガイド-運転初心者
安全に右折するための完全ガイド | 車を上手く運転しちゃおう!

Var(X-Y)

Var: variance
Cov: covariance

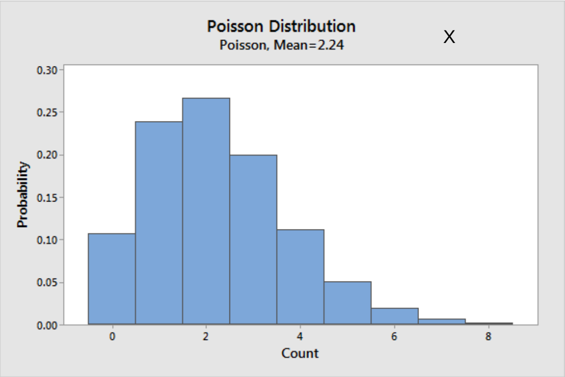
- Var(X+Y)= Var(X) + Var(Y) + Cov(X,Y) [恆等式(永遠成立)]
  - 假如Cov(X,Y) = 0 (例如，兩者獨立)則Var(X+Y)= Var(X) + Var(Y)

- Var(X-Y) = Var(X) + Var(Y) – Cov(X,Y) [恆等式(永遠成立)]
  - 假如Cov(X,Y) = 0 (例如，兩者獨立)則Var(**X-Y**)= Var(X) + Var(Y)

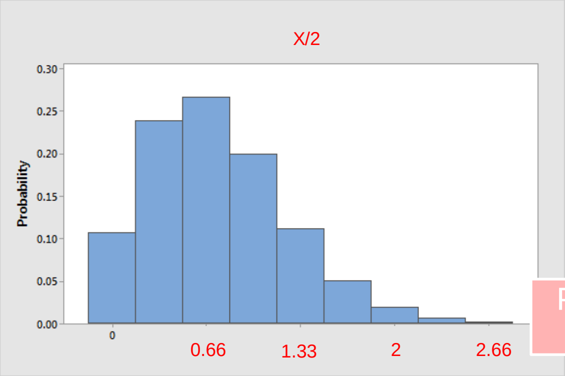
朴瓦松縮放後不是朴瓦松

https://statisticsbyjim.com/probability/poisson-distribution/

- 朴瓦松 X ~ Poi(λ)
  - 例如：X ~ Poi(λ = 2.24)



- 縮放後
  - 例如： X/3 很顯然不是朴瓦松



Poisson不會有小數點

Ordinal Data & others

- 四種資料尺度

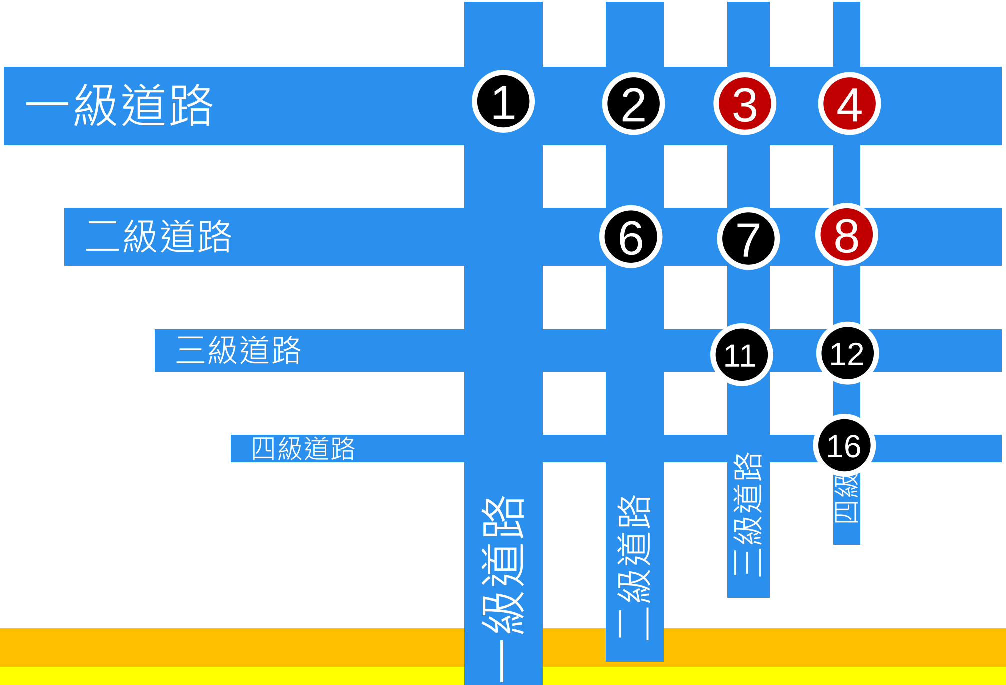
Once the value of a variable has been determined, the calculations can be carried out. However, variables are not the same and it is not possible to perform the same calculations for all variables. For example, there is an average 24-hour traffic volume, but there is no such thing as an average mode of transport. This has to do with the level at which the variable is measured. There are four types of measurement level or scale:

- nominal;
- ordinal;
- interval;
- ratio.

C.R.O.W. p.152

Ordinal

- Ordinal有順序關係，但是尺度之間並不等距
- 例如：用一個雙元變數κ，辨認跨兩級道路



$\kappa_1 = 0;$
$\kappa_2 = 0;$
$\kappa_3 = 1;$
$\kappa_4 = 1;$
$\kappa_6 = 0;$
$\kappa_7 = 0;$
$\kappa_8 = 1;$
$\kappa_{11} = 0;$
$\kappa_{12} = 0;$
$\kappa_{16} = 0;$

(續)

- 但如果認為道路層級是ordinal，可以再額外細部修正
  - 基準:第一級與第三級跨
  - 第一級與第四級跨 ω
  - 第二級與第四級跨 γ

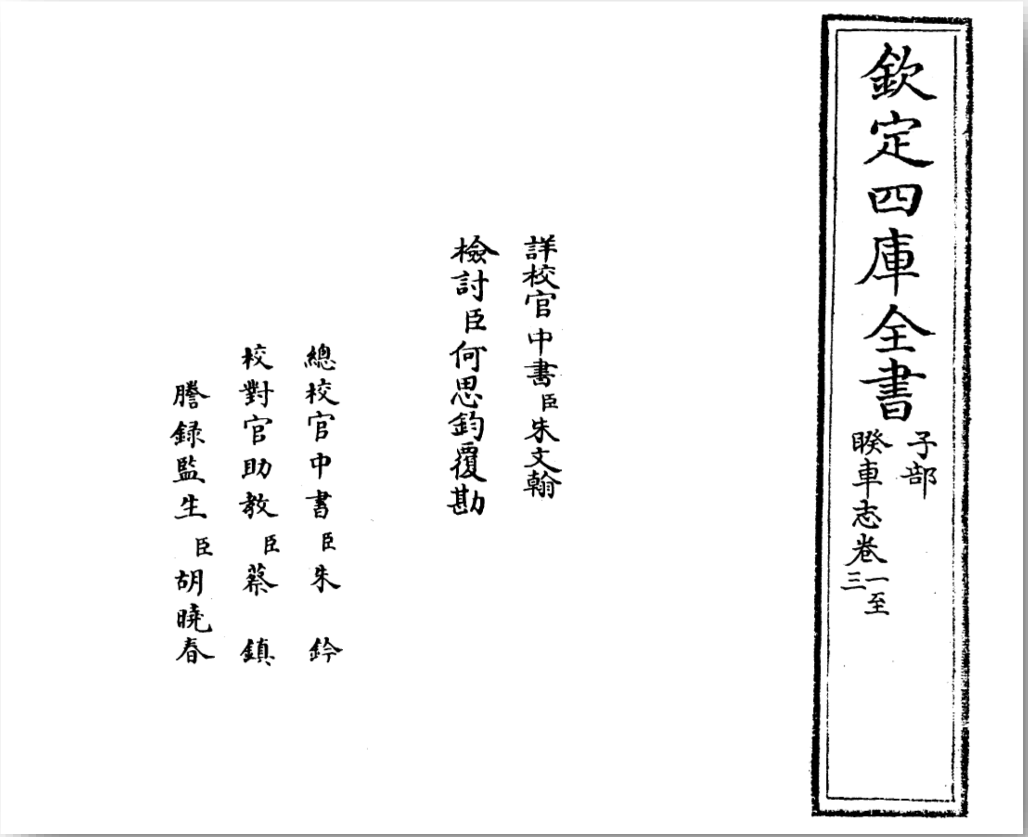
$\kappa_1 = 0;$
$\kappa_2 = 0;$
$\kappa_3 = 1;$
$\kappa_4 = 1;$
$\kappa_6 = 0;$
$\kappa_7 = 0;$
$\kappa_8 = 1;$
$\kappa_{11} = 0;$
$\kappa_{12} = 0;$
$\kappa_{16} = 0;$

$\omega_4 = \omega_{13} = 1;$
$\omega_i = 0$ otherwise

$\gamma_8 = \gamma_{14} = 1;$
$\gamma_i = 0$ otherwise

《睽車志》

• 《睽車志》收錄於[清] 四庫全書


欽定四庫全書
子部
睽車志卷一至三

詳校官中書臣朱文翰
檢討臣何思鈞覆勘

總校官中書臣朱鈐
校對官助教臣蔡鎮
謄錄監生臣胡曉春

睽車志　　小説家類二異聞之屬

提要

臣等謹案睽車志六卷宋郭彖撰彖字次象和州人由進士歴官知興國軍是書皆紀鬼怪神異之事為當時耳目所聞見者其名睽車志蓋取睽上六載鬼一車之語也宋史藝文志載有是書一卷陳振孫書錄解題馬端

臨經籍考俱作五卷而明會稽商濬刻入稗海者又作六卷參錯不一蓋後人屢有所分析故卷目多寡互異耳書中所載多建炎紹興乾道淳熙間事而汴京舊聞亦間為錄入各條之末悉分注某人所説蓋用洪邁夷堅志之例其大旨亦主於因果報應以為世勸戒特摭拾既廣有聞即錄往往傳述附會於事實不免荒唐如米芾本北宋名流而疑為

紀昀(紀曉嵐)

鱗精程迴亦南渡後儒宿多所著述而以為
其家奉玉真娘子由此致富張觷能斥奸平
亂志操甚正身後至廟食邵武而以為挾嫌
殺人白晝見鬼而卒此類皆灼然可知其妄
其他紀載亦多涉瑣碎然齊諧志怪之書古
所不廢雖無裨考證而有助談資此書亦搜
神記稽神錄之支流故仍錄而存之以備小
說家之一種焉乾隆四十二年六月恭校上

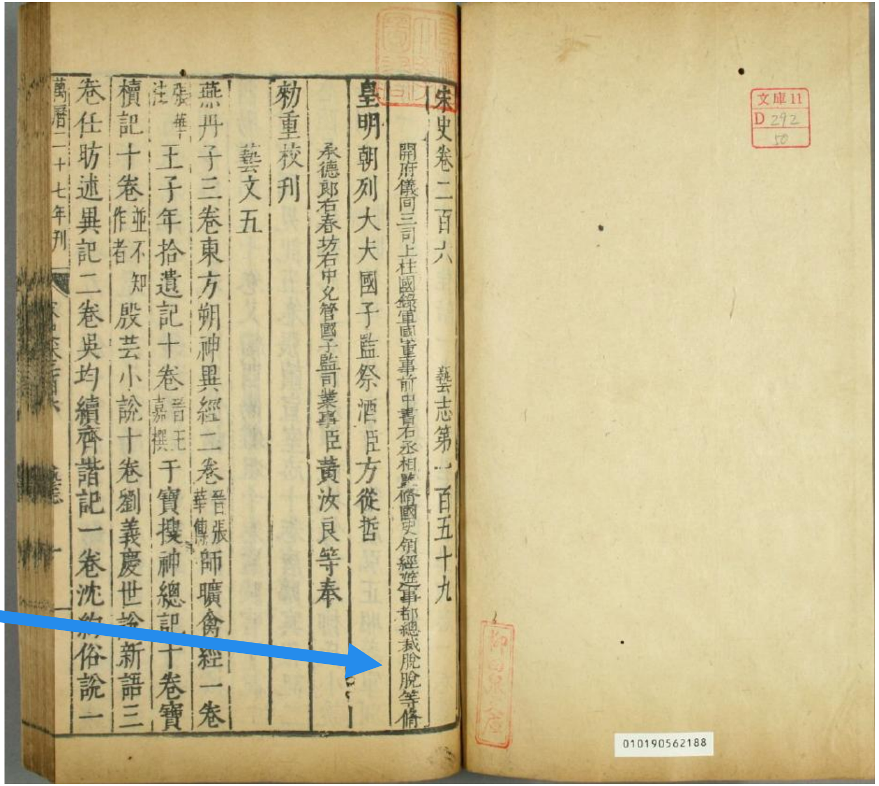
總纂官臣紀昀臣陸錫熊臣孫士毅
總校官臣陸費墀

《睽車志》

- 《睽車志》撰於宋
  - [元]脫脫 主編《宋史藝文志》卷五，提及此書，稱有一卷

脫脫(托克托)

資料來源：《奇皇后》劇照


宋史卷二百六 藝文志第一百五十九
開府儀同三司上柱國錄軍國重事前中書右丞相監修國史領經筵事都總裁脫脫等修
皇明朝列大夫國子監祭酒臣方從哲
承德郎右春坊右中允管國子監司業事臣黃汝良等奉
勅重校刊
藝文五
燕丹子三卷 東方朔神異經二卷 張華注 師曠禽經一卷 晉張華傳
王子年拾遺記十卷 晉王嘉撰 于寶搜神總記十卷 寶
積記十卷 並不知作者 殷芸小說十卷 劉義慶世說新語三
卷 任昉述異記二卷 吳均續齊諧記一卷 沈約俗說一
萬曆二十七年刊

天與仲之所纂不著姓張氏儆誡會最一卷唯室先生步里客談一卷沈括筆談二十五卷又清夜錄一卷王銍續清夜錄一卷郭彖睽車志一卷洪邁隨筆五集七十四卷又夷堅志六十卷甲乙丙志夷堅志八十卷丁戊己庚志胡仔漁隱叢話前後集四十卷姚逈隨因紀述一卷王煥北山紀事十二卷何晦摭言十五卷又廣摭言十五卷僧贊寧傳載八卷徐鉉稽神錄十卷蘇轍龍川志六卷蘇軾東坡詩話一卷楊囦道四六餘話二卷謝伋四六談麈二卷葉凱南宮詩話一卷葉夢得石林避暑錄二卷馬永卿懶真子五卷趙槩見聞錄二卷王同叙事一卷劉斧

翰府名談二十五卷又摭遺二十卷青瑣高議十八卷僧文瑩湘山野錄三卷又玉壺清話十卷李端彥賢已集三十二卷王陶談淵一卷錢明逸衣冠盛事一卷句頵坐右書一卷曾鞏雜職一卷張師正怪集五卷又倦遊雜錄十二卷括異志十卷畢仲詢幕府燕閑錄十卷劉攽三異記一卷岑象求吉凶影響錄八卷麗元英南齋雜錄一卷孔平仲釋稗一卷又續世說十二卷孔氏雜說一卷魏泰訂誤集二卷又東軒筆錄十五卷陳正敏遯溪野話三卷又遯齋閑覽十四卷李薦師友談記十卷王山筆奩錄七卷董逌錢譜十卷王闢之澠水燕談十卷

經十卷郭璞序不著姓名袁天剛玄成子一卷孫思邈坐照論
弁五行法一卷柳淸風周世明等玉冊寶文八卷李淳
風立觀經一卷僧一行地理經十五卷呼龍經一卷金
歌四季氣色訣一行撰論孫知古人倫龜鑑三卷王澄陰陽
二宅集要二卷僧正固骨法明鏡三卷丘延翰銅函記
一卷天定盤古局一卷漢赤松子海角經一卷明鏡碎
金七卷唐舉肉眼通神論三卷金鑠歌一卷鬼谷子觀
氣色出圖相一卷黃石公八宅二卷許負相訣三卷李
淳風一行禪師塟律祕密經十卷呂才楊烏子改墳枯
骨經一卷曾楊一青囊經歌二卷楊救貧正龍子經一

卷曾文展八分歌一卷李筌金華經三卷宋齊丘玉管
照神局二卷天花經三卷序云黃巢得於長安晏氏辨氣色上面
詩一卷不知名劉虛白三輔學堂正訣一卷危延真相法
一卷五星六曜面部訣一卷裴仲卿玄珠囊骨法一卷
劉度具氣色真相法一卷王希逸地理祕妙歌訣一卷
地理名山異形歌一卷孫臏塟白骨曆卷亡隱逸人玉環
經一卷不知姓名天涯海角經一卷不知作者九江李麟注解徽宗太平
曆覽圖一卷陳摶人倫風鑑一卷司空班范越鳳尋龍
入式歌一卷王洙地理新書三十卷蘇粹明地理指南
三卷蔡望五家通天局一卷報應九星妙術文局一卷

論地理一卷老子地鑑訣祕術一卷五姓合諸家風水
地理一卷昭幽記一卷鬼靈經拜枯骨經二卷唐刪定
陰陽葬經二卷唐書地理經十卷青烏子歌訣二卷金
鷄曆一卷五音二十八將圖一卷赤松子三卷易括地
林一卷丘延翰五家通天局一卷夫子掘斗記一卷孔
子金鑾記一卷推背圖一卷鬼谷子白虎經一卷又白
虎五通經訣一卷洞幽祕要圖一卷孝經雌雄圖四卷
河上公金藏祕訣要略一卷玄珠握鑑三卷玉函寶鑑
三卷真人水鑑十三卷張華三鑑靈書三卷陶弘景握
鑑方三卷證應集三卷金婁先生祕訣三卷真圖祕訣

一卷銘軌五卷胡濟川小遊七十二局立成一卷大小
遊三奇五福立成一卷十一神旁通太歲甲子圖一卷
曹植黃帝寶藏經三卷括明經一卷悟迷經一卷余考
一作秀且暮經一卷神樞萬一祕經一卷紀重政祕要一
卷雷公印法三卷雷公撮殺律一卷徐遂發蒙一卷玄
女十課一卷呂佐周地論七曜一卷陰術氣神一卷七
曜氣神歷代帝紀五卷玉堂祕訣一卷大運祕要心髓
訣一卷呂才陰陽書一卷五姓鳳髓寶鑑論一卷陰陽
雜要一卷玄珠錄要三卷張良玄悟歌一卷斗書一卷
陰陽二卷論一卷黃帝四序經一卷寶臺七賢論一卷

禮諸史提要十五卷陳傅良漢兵制一卷備邊十策九
卷徐天麟西漢會要七十卷漢兵本末一卷曾慥類說
五十卷錢文子補漢兵志一卷錢諷史韻四十二卷鄭
應龍務學須知二卷高似孫緯略十二卷子略四卷吳
曾南北分門事類十二卷魏彥惇名臣四科事實十四
卷王掄羣玉義府五十四卷范師道垂拱元龜會要詳
節四十卷國朝類要十二卷俞鼎俞經儒學警悟四十
卷鄭厚通鑑分門類要四十卷柳正夫西漢蒙求一卷
李孝美文房監古三卷竇苹載籍討源一卷王仲閎語
本二十五卷毛友左傳類對賦六卷俞觀能孝經類鑑

七卷胡宏叙古蒙求一卷玉山題府二十卷熙寧題髓
十五卷帝王事實十卷聖賢事實十卷漢唐事實十五
卷國朝韻對八卷引證事類三十卷曾史分門屬類賦
一卷古今通編八卷諸子談論三卷並不知作者
右類事類三百七部一萬一千三百九十三卷
黃帝內經素問二十四卷唐王冰注素問八卷隋全元起注黃帝
靈樞經九卷黃帝鍼經九卷黃帝灸經明堂三卷黃帝
九虛內經五卷楊玄操素問釋音言一作一卷素問醫療
訣一卷秦越人難經疏十三卷黃帝脉經一卷又脉訣
一卷張仲景脉經一卷又五藏榮衛論一卷耆婆脉經

三卷徐氏脉經三卷王叔和脉訣一卷一作經一卷孩子脉論
一卷李勣脉經一卷張及脉經手訣一卷王善主徐裔脉
訣二卷韓氏脉訣一卷脉經一卷百會要訣脉經一卷
碎金脉訣一卷元門脉訣一卷身經要集一卷大醫秘
訣診候生死部一卷倉公決死生秘要一卷神農五藏
論一卷黃帝五藏論一卷黃庭五藏經一卷黃庭五藏
六府圖一卷趙業黃庭五藏論七卷張向容大五藏論
一卷又小五藏論一卷五藏金鑑論一卷段元亮一作允
五藏鑑元一作原四卷孫思邈五藏旁通明鑑圖一卷又
針經一卷張文懿藏府通玄賦一卷五藏攝養明鑑圖

一卷吳兢五藏論應家象一卷裴王庭五色旁通五藏
圖一卷五藏要訣一卷太元心論一卷岐伯針經一卷
扁鵲鍼傳一卷玄悟四神針經一卷甄權針經抄三卷
王處明玄秘會要針經五卷呂博金滕玉匱鍼經三卷
黃帝問岐伯灸經一卷顏齊　灸經十卷明堂灸法三
卷皇甫謐黃帝三部鍼灸經十二卷即甲乙經岐伯論針灸
要訣一卷山眺一作兆鍼灸經一卷公孫克針灸經一卷
吳復珪小兒明堂針灸經一卷王惟一明堂經三卷明
堂玄真經訣一卷朱遂明堂論一卷金鑑集歌一卷黃
帝大素經三卷楊上善注刺法一卷太上天寶金鏡靈樞神

景內編九卷扁鵲注黃帝八十一難經二卷秦越人撰扁鵲
脈經一卷張仲景傷寒論十卷五藏論一卷王叔和脈
經十卷脈訣機要三卷巢元方巢氏諸病源候論五十
卷崔知悌灸勞法一卷王冰素問六脈玄珠密語一卷
褚澄褚氏遺書一卷華佗藥方一卷金匱要略方三卷
張仲景撰王叔和集葛洪肘後備急百一方三卷劉涓子神仙遺
論十卷東蜀李頓錄宇文士及粧臺記六卷師巫顱顖經二
卷孫思邈千金方三十卷千金髓方二十卷千金翼方
三十卷玉函方三卷王起仙人水鏡一卷王燾外臺秘
方四十卷陳藏器本草拾遺十卷孔志約唐本草二十

卷李昉開寶本草二十卷目一卷盧多遜詳定本草二
十卷目錄一卷補注本草二十卷目錄一卷李含光本
草音義五卷蕭炳四聲本草四卷本草韻略五卷楊損
之刪繁本草五卷杜善方本草性類一卷陳士良食性
本草十卷龐安時難經解義一卷朱庭臣黃帝八十一
難經注釋一卷張仲景療黃經一卷又口齒論一卷金
匱玉函八卷王叔和集扁鵲療黃經三卷又枕中秘訣三卷
靑烏子風經一卷吳希言風論山兆一作眺經一卷文義
方通玄經十卷呂廣金韜玉鑑經二卷雷一作靈公仙人
養性治理一作身經三卷醫源兆經一卷林億黃帝三部